

Chapter 16

Limits and Derivatives

Solutions

SECTION - A

Objective Type Questions (One option is correct)

1. Let $\lim_{x \rightarrow 0} \frac{\sin 2x}{x} = a$ and $\lim_{x \rightarrow 0} \frac{3x}{\tan x} = b$, then $a + b$ equals

- (1) 5 (2) 6 (3) 0 (4) 4

Sol. Answer (1)

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x} = \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right) \cdot 2 = 2$$

$$\Rightarrow a = 2$$

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{\tan x} = 3 \lim_{x \rightarrow 0} \left(\frac{x}{\tan x} \right) = 3$$

$$\Rightarrow b = 3$$

$$a + b = 2 + 3 = 5$$

2. Let $\lim_{x \rightarrow 0} \frac{\sin 2x}{\tan\left(\frac{x}{2}\right)} = L_1$ and $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} = L_2$, then the value of $L_1 L_2$ is

- (1) 4 (2) 8 (3) 6 (4) 2

Sol. Answer (2)

$$L_1 = \lim_{x \rightarrow 0} \frac{\left(\frac{\sin 2x}{2x} \right) \cdot 2x}{\left(\frac{\tan\left(\frac{x}{2}\right)}{\left(\frac{x}{2}\right)} \right) \cdot \frac{x}{2}} = 4$$

$$L_2 = \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} = \lim_{x \rightarrow 0} \left(\frac{e^{2x} - 1}{2x} \right) \cdot 2 = 2$$

$$\Rightarrow L_1 L_2 = 8$$

3. $\lim_{x \rightarrow 0} \frac{2^x - 1}{x} + \lim_{x \rightarrow 0} \frac{3^x - 1}{x} - \lim_{x \rightarrow 0} \left(\frac{6^x - 1}{x} \right)$ equals

(1) 1

(2) 2

(3) 0

(4) 4

Sol. Answer (3)

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{2^x - 1}{x} + \lim_{x \rightarrow 0} \frac{3^x - 1}{x} - \lim_{x \rightarrow 0} \frac{6^x - 1}{x} \\ &= \log 2 + \log 3 + \log 6 \\ &= \log \left(\frac{6}{6} \right) = \log 1 = 0 \end{aligned}$$

4. $\lim_{x \rightarrow 0} \frac{\log(1+2x)}{x} + \lim_{x \rightarrow 2} \frac{x^4 - 2^4}{x - 2}$ equals

(1) 30

(2) 32

(3) 35

(4) 34

Sol. Answer (4)

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\log(1+2x)}{x} + \lim_{x \rightarrow 2} \frac{x^4 - 2^4}{x - 2} \\ &= \lim_{x \rightarrow 0} \left(\frac{\log(1+2x)}{2x} \right) \cdot 2 + \lim_{x \rightarrow 2} 4 \cdot x^3 \\ &= 2 + 32 = 34 \end{aligned}$$

5. $\lim_{x \rightarrow \infty} \frac{\sin x}{x} + \lim_{x \rightarrow \infty} \frac{\log x}{x}$ equals

(1) 0

(2) 1

(3) 3

(4) ∞ **Sol.** Answer (1)

$$\lim_{x \rightarrow \infty} \frac{\sin x}{x} + \lim_{x \rightarrow \infty} \frac{\log x}{x} = 0 + 0 = 0$$

6. $\lim_{x \rightarrow \infty} (\sqrt{x+1} - \sqrt{x})$ equals

(1) ∞

(2) 0

(3) -1

(4) 1

Sol. Answer (2)

$$\begin{aligned} & \lim_{x \rightarrow \infty} \sqrt{x+1} - \sqrt{x} \\ &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x+1} - \sqrt{x})(\sqrt{x+1} + \sqrt{x})}{\sqrt{x+1} + \sqrt{x}} \\ &= \lim_{x \rightarrow \infty} \frac{(x+1-x)}{\sqrt{x+1} + \sqrt{x}} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x+1} + \sqrt{x}} \\ &= \frac{1}{\infty} = 0 \end{aligned}$$

7. $\lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1} =$

(1) 2

(2) $\log_e 2$

(3) $\frac{\log_e 2}{2}$

(4) $2 \log_e 2$

Sol. Answer (4)

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1} \quad \text{from } \left(\frac{0}{0} \right)$$

Using L' Hospital Rule, we get

$$\lim_{x \rightarrow 0} \frac{2^x \log 2}{\frac{1}{2}(1+x)^{-1/2}} = 2 \log 2$$

Aliter :

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1} = \lim_{x \rightarrow 0} \frac{2^x - 1}{x} \times \frac{x}{\sqrt{1+x} - 1} \times \frac{\sqrt{1+x} + 1}{\sqrt{1+x} + 1} = \log 2 \lim_{x \rightarrow 0} (\sqrt{1+x} + 1) = 2 \log 2$$

8. $\lim_{x \rightarrow 1} \left(\sec \frac{\pi}{2^x} \right) \cdot \ln x$ is equal to

(1) $\frac{\pi}{2}$

(2) $-\frac{2}{\pi \ln 2}$

(3) $\frac{2}{\pi \ln 2}$

(4) $\frac{\pi}{\ln 2}$

Sol. Answer (3)

$$\lim_{x \rightarrow 1} \sec \frac{\pi}{2^x} \cdot \ln x$$

$$= \lim_{x \rightarrow 1} \frac{\ln x}{\cos \frac{\pi}{2^x}}$$

$$= \lim_{x \rightarrow 1} \frac{\frac{1}{x}}{(-)\pi \sin \frac{\pi}{2^x} \cdot \frac{1}{2^x} \ln 2(-1)}$$

$$= \frac{1}{\pi \sin \frac{\pi}{2} \times \frac{1}{2} \times \ln 2}$$

$$= \frac{2}{\pi \ln 2}$$

9. $\lim_{x \rightarrow 0} \frac{\sin 2x}{2 - \sqrt{4-x}}$ is

(1) 2

(2) 4

(3) 8

(4) 0

Sol. Answer (3)

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{2 - \sqrt{4 - x}} \times \frac{(2 + \sqrt{4 - x})}{(2 + \sqrt{4 - x})}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin 2x \times (2 + \sqrt{4 - x})}{(4 - 4 + x)}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin 2x \times 2 + \sqrt{4 - x}}{\frac{2x}{2}}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2 + \sqrt{4 - x}}{\frac{1}{2}}$$

$$\Rightarrow 8$$

10. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x\sqrt{x^2}}$ equals to

(1) $\frac{1}{2}$

(2) $-\frac{1}{2}$

(3) 0

(4) Does not exist

Sol. Answer (4)

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x |x|}$$

Now,

$$\lim_{x \rightarrow 0^-} \frac{2 \sin^2 \frac{n}{2}}{-x^2}$$

$$\Rightarrow \lim_{x \rightarrow 0^-} -\frac{2 \sin^2 \frac{x}{2}}{4 \times \frac{x^2}{4}}$$

$$\Rightarrow -\frac{1}{2}$$

$$\text{and } \lim_{x \rightarrow 0^+} \frac{2 \sin^2 \frac{x}{2}}{+x^2}$$

$$\Rightarrow \frac{1}{2}$$

Does not exist.

11. $\lim_{x \rightarrow 0} ((x \cot x) + (x \log x))$ equals

(1) 1

(2) 2

(3) 3

(4) 0

Sol. Answer (1)

$$\lim_{x \rightarrow 0} x \cot x + \lim_{x \rightarrow 0} x \log x$$

$$= \lim_{x \rightarrow 0} \frac{x}{\tan x} + \lim_{x \rightarrow 0} \frac{\log x}{\left(\frac{1}{x}\right)}$$

$$= \lim_{x \rightarrow 0} \frac{x}{\tan x} + \lim_{x \rightarrow 0} \frac{\left(\frac{1}{x}\right)}{\left(-\frac{1}{x^2}\right)} = 1 + \lim_{x \rightarrow 0} (-x)$$

$$= 1 + 0 = 1$$

12. The value of $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right)$ equals

(1) 1

(2) -1

(3) 2

(4) 0

Sol. Answer (4)

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\tan x} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\tan x - x}{x \tan x}$$

$$= \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{x \sec^2 x + \tan x}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sec x \sec x \tan x}{x(2 \sec x \sec x \tan x) + \sec^2 x + \sec^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{0}{2} = 0$$

13. $\lim_{x \rightarrow -\infty} \left(\frac{x^4 \sin\left(\frac{1}{x}\right) + x^2}{1 + |x|^3} \right)$ equals

(1) -1

(2) 1

(3) 0

(4) 2

Sol. Answer (1)

$$\begin{aligned}
 & \lim_{x \rightarrow -\infty} \left(\frac{x^4 \sin\left(\frac{1}{x}\right) + x^2}{1 + |x|^3} \right) \\
 &= \lim_{x \rightarrow -\infty} \left(\frac{x^4 \sin\left(\frac{1}{x}\right) + x^2}{1 - x^3} \right) \\
 &= \lim_{x \rightarrow -\infty} \frac{x^2 \sin\left(\frac{1}{x}\right) + 1}{\frac{1}{x^2} - x} \\
 &= \lim_{x \rightarrow -\infty} \frac{x \sin\left(\frac{1}{x}\right) + \frac{1}{x}}{\frac{1}{x^3} - 1} \\
 &= \lim_{x \rightarrow -\infty} \frac{\left(\frac{\sin\left(\frac{1}{x}\right)}{\left(\frac{1}{x}\right)} + \frac{1}{x} \right)}{\frac{1}{x^3} - 1} = -1
 \end{aligned}$$

14. $m, n \in I^+$, then $\lim_{x \rightarrow 0} \frac{\sin x^n}{(\sin x)^m}$ equals

- (1) 1, if $n < m$ (2) 0, if $n = m$ (3) $\frac{n}{m}$ (4) 0, if $n > m$

Sol. Answer (4)

Given $m, n \in I^+$

$$\text{Now, } \lim_{x \rightarrow 0} \frac{\sin x^n}{(\sin x)^m} = \lim_{x \rightarrow 0} \frac{\left(\frac{\sin x^n}{x^n} \right) x^n}{\left(\frac{\sin x}{x} \right)^m \cdot x^m} = \lim_{x \rightarrow 0} x^{n-m} = 0 \text{ for } n > m.$$

15. $\lim_{x \rightarrow \infty} \frac{(1+x+x^2)}{(\ln x)^3}$ is equal to

- (1) 2 (2) e^2 (3) e^{-2} (4) ∞

Sol. Answer (4)

$$\lim_{x \rightarrow \infty} \frac{1+x+x^2}{(\log x)^3}$$

$$\lim_{x \rightarrow \infty} \frac{1+2x}{3(\ln x)^2} \cdot \frac{1}{x} = \lim_{x \rightarrow \infty} \frac{x+2x^2}{3(\ln x)^2} = \frac{1}{3} \lim_{x \rightarrow \infty} \frac{(1+4x)x}{2(\ln x)} = \frac{1}{6} \lim_{x \rightarrow \infty} \frac{1+8x}{\frac{1}{x}} = \infty$$

16. For $x \in R$, $\lim_{x \rightarrow \infty} \left(\frac{x-3}{x+2} \right)^x$ equals

(1) e (2) e^{-1} (3) e^{-5} (4) e^5 **Sol. Answer (3)**

$$\lim_{x \rightarrow \infty} \left(\frac{x-3}{x+2} \right)^x = e^{\lim_{x \rightarrow \infty} \left(\frac{x-3}{x+2} - 1 \right)x} = e^{\lim_{x \rightarrow \infty} \frac{-5x}{x+2}} = e^{\lim_{x \rightarrow \infty} \frac{-5}{1+\frac{2}{x}}} = e^{-5}$$

17. $\lim_{x \rightarrow 0} \left(\frac{4^x + 9^x}{2} \right)^{\frac{1}{x}}$ equals

(1) 2

(2) 6

(3) 16

(4) 112

Sol. Answer (2)

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{4^x + 9^x}{2} \right)^{\frac{1}{x}} &= e^{\lim_{x \rightarrow 0} \left(\frac{4^x + 9^x}{2} - 1 \right) \cdot \frac{1}{x}} = e^{\lim_{x \rightarrow 0} \left(\frac{4^x + 9^x - 2}{2x} \right)} \\ &= e^{\lim_{x \rightarrow 0} \frac{4^x \log 4 + 9^x \log 9}{2}} \\ &= e^{\left(\frac{\log 4 + \log 9}{2} \right)} = e^{\log_6 6} = 6 \end{aligned}$$

18. If $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2} \right)^{2x} = e^2$, then the values of a and b are

(1) $a \in R, b \in R$ (2) $a = 1, b \in R$ (3) $a \in R, b = 2$ (4) $a = 1, b = 2$ **Sol. Answer (2)**

$$\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2} \right)^{2x} = e^2 \Rightarrow e^{\lim_{x \rightarrow \infty} \left(\frac{a}{x} + \frac{b}{x^2} \right) \cdot 2x} = e^2$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{a}{x} + \frac{b}{x^2} \right) \cdot 2x = 2 \Rightarrow \lim_{x \rightarrow \infty} 2x \left(\frac{a}{x} + \frac{b}{x^2} \right) = 2$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(2a + \frac{2b}{x} \right) = 2 \Rightarrow a = 1, b \in R$$

19. If α, β are roots of $ax^2 + bx + c = 0$, then $\lim_{x \rightarrow \alpha} (1 + ax^2 + bx + c)^{\frac{1}{x-\alpha}}$ is

- (1) $a(\alpha - \beta)$ (2) $\log e^{a(\alpha - \beta)}$ (3) $e^{a(\alpha - \beta)}$ (4) $e^{a(\alpha + \beta)}$

Sol. Answer (3)

$$\begin{aligned} & \lim_{x \rightarrow \alpha} (1 + ax^2 + bx + c)^{\frac{1}{x-\alpha}} \\ &= e^{\lim_{x \rightarrow \alpha} (ax^2 + bx + c) \cdot \frac{1}{(x-\alpha)}} \\ &= e^{\lim_{x \rightarrow \alpha} a(x-\alpha)(x-\beta) \cdot \frac{1}{(x-\alpha)}} \\ &= e^{a(\alpha - \beta)} \end{aligned}$$

20. If $\lim_{x \rightarrow 0} (\cos x + a \sin bx)^{\frac{1}{x}} = e^2$, then the values of a and b are

- (1) $a = 1, b = -2$ (2) $a = 2\sqrt{2}, b = \sqrt{2}$ (3) $a = 2\sqrt{2}, b = \frac{1}{\sqrt{2}}$ (4) $a = -2, b = 1$

Sol. Answer (3)

$$\begin{aligned} & \lim_{x \rightarrow 0} (\cos x + a \sin bx)^{\frac{1}{x}} = e^2 \\ \Rightarrow & \lim_{x \rightarrow 0} (\cos x + a \sin bx - 1) \left(\frac{1}{x} \right) = 2 \\ \Rightarrow & \lim_{x \rightarrow 0} \frac{\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) + a \left(bx - \frac{(bx)^3}{3!} + \frac{(bx)^5}{5!} - \dots \right) - 1}{x} = 2 \\ \Rightarrow & \lim_{x \rightarrow 0} \frac{abx + x^2 \left(-\frac{1}{2!} \right) + x^3 \left(-\frac{ab^3}{3!} \right) + \dots}{x} = 2 \\ \Rightarrow & ab = 2 \end{aligned}$$

Hence option (3) is correct.

21. Let α and β be real and distinct root of $ax^2 + bx + c = 0$ then $\lim_{x \rightarrow \alpha} [1 + \sin(ax^2 + bx + c)]^{\frac{1}{x-\alpha}}$ is

- (1) $e^{\alpha - \beta}$ (2) $e^{\beta - \alpha}$ (3) $e^{a(\alpha - \beta)}$ (4) $e^{a(\beta - \alpha)}$

Sol. Answer (3)

$$\begin{aligned} & \lim_{x \rightarrow \alpha} (1 + \sin(ax^2 + bx + c))^{\frac{1}{x-\alpha}} \\ &= e^{\lim_{x \rightarrow \alpha} \left(\frac{\sin(ax^2 + bx + c)}{x-\alpha} \right)} = e^{\lim_{x \rightarrow \alpha} \left(\frac{\sin(a(x-\alpha)(x-\beta))}{(x-\alpha)} \right)} \\ &= e^{\lim_{x \rightarrow \alpha} \left(\frac{\sin a(x-\alpha)(x-\beta)}{a(x-\alpha)(x-\beta)} \right) \left(\frac{a(x-\alpha)(x-\beta)}{(x-\alpha)} \right)} = e^{a(\alpha-\beta)} \end{aligned}$$

22. Let $f(2) = 4$, $f'(2) = 4$. Then $\lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x-2}$ is

(1) $-\frac{1}{3}$

(2) -2

(3) -4

(4) 3

Sol. Answer (3)

It is given that $f(2) = 4$ and $f'(2) = 4$,

$$\text{Now, } \lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x-2} = \lim_{x \rightarrow 2} \frac{f(2) - 2f'(x)}{1} = f(2) - 2f'(2) = -4$$

23. If $f(4) = 4$, $f'(4) = 1$, then $\lim_{x \rightarrow 4} \frac{2 - \sqrt{f(x)}}{2 - \sqrt{x}}$ is equal to

(1) 0

(2) 1

(3) -1

(4) 2

Sol. Answer (2)

Given $f(4) = 4$, $f'(4) = 1$

$$\text{Now, } \lim_{x \rightarrow 4} \frac{2 - \sqrt{f(x)}}{2 - \sqrt{x}} = \lim_{x \rightarrow 4} \frac{-\frac{1}{2\sqrt{f(x)}} f'(x)}{-\frac{1}{2\sqrt{x}}} = \frac{f'(4)}{\sqrt{f(4)}} \cdot \sqrt{4} = \frac{1}{2} \times 2 = 1$$

24. If $f(x)$ and $g(x)$ be differentiable functions and $f(1) = g(1) = 2$ then $\lim_{x \rightarrow 1} \frac{f(1) \cdot g(x) - f(x) \cdot g(1) - f(1) + g(1)}{g(x) - f(x)}$ is equal to

(1) 0

(2) 1

(3) 2

(4) -1

Sol. Answer (3)

Given $f(1), g(1) = 2$

$$\text{Let } y = \lim_{x \rightarrow 1} \frac{f(1) \cdot g(x) - f(x) \cdot g(1) - f(1) + g(1)}{g(x) - f(x)}$$

By putting $g(1) = f(1) = 2$, we get

$$y = \lim_{x \rightarrow 1} \frac{2g(x) - 2f(x)}{g(x) - f(x)} = \lim_{x \rightarrow 1} \frac{2[g(x) - f(x)]}{g(x) - f(x)} = 2$$

25. Let α and β be the distinct roots of $ax^2 + bx + c = 0$, then $\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$ is equal to

- (1) $\frac{1}{2}(\alpha - \beta)^2$ (2) $-\frac{a^2}{2}(\alpha - \beta)^2$ (3) 0 (4) $\frac{a^2}{2}(\alpha - \beta)^2$

Sol. Answer (4)

Since α, β are the roots of $ax^2 + bx + c = 0$

$$\therefore ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

$$\text{Now, } \lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$$

$$= \lim_{x \rightarrow \alpha} \frac{2\sin^2 \frac{1}{2}(ax^2 + bx + c)}{(x - \alpha)^2}$$

$$= \lim_{x \rightarrow \alpha} \frac{2\sin^2 \left(\frac{1}{2}a(x - \alpha)(x - \beta) \right)}{(x - \alpha)^2}$$

$$= \lim_{x \rightarrow \alpha} \frac{2 \left(\sin \left(\frac{a}{2}(x - \alpha)(x - \beta) \right) \right)^2}{\left(\frac{a}{2}(x - \alpha)(x - \beta) \right)^2} \cdot \left(\frac{a}{2}(x - \alpha)(x - \beta) \right)^2 \cdot \frac{1}{(x - \alpha)^2}$$

$$= \lim_{x \rightarrow \alpha} \frac{2 \cdot \frac{a^2}{4}(x - \alpha)^2(x - \beta)^2}{(x - \alpha)^2}$$

$$= \frac{a^2}{2}(\alpha - \beta)^2$$

26. If α and β be the roots of the equation $ax^2 + bx + c = 0$ then $\lim_{x \rightarrow \frac{1}{\alpha}} \sqrt{\frac{1 - \cos^2(cx^2 + bx + a)}{4(1 - \alpha x)^2}}$

- (1) Does not exist (2) Equals $\left| \frac{c}{2\alpha} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) \right|$ (3) Equals $\left| \frac{c}{2\alpha} \left(\frac{1}{\alpha} - \frac{1}{\beta} \right) \right|$ (4) Equals $\left| \frac{c}{2} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) \right|$

Sol. Answer (3)

$\therefore \alpha, \beta$ are root of $ax^2 + bx + c = 0$

$\therefore \frac{1}{\alpha}, \frac{1}{\beta}$ are roots of $cx^2 + bx + a = 0$

$$\begin{aligned}
 \text{Now, } \lim_{x \rightarrow \frac{1}{\alpha}} \sqrt{\frac{1 - \cos^2(cx^2 + bx + a)}{4(1 - \alpha x)^2}} \\
 &= \lim_{x \rightarrow \frac{1}{\alpha}} \frac{|\sin(cx^2 + bx + a)|}{2|1 - \alpha x|} \\
 &= \lim_{x \rightarrow \frac{1}{\alpha}} \left| \frac{\sin(cx^2 + bx + a)}{(cx^2 + bx + a)} \cdot \frac{(cx^2 + bx + a)}{2|1 - \alpha x|} \right| \\
 &= \lim_{x \rightarrow \frac{1}{\alpha}} \left| \frac{cx^2 + bx + a}{2(1 - \alpha x)} \right|
 \end{aligned}$$

$$\text{Let } y = \frac{1}{x} \Rightarrow y \rightarrow \alpha$$

$$\lim_{x \rightarrow \frac{1}{\alpha}} \left| \frac{cx^2 + bx + a}{2(1 - \alpha x)} \right|$$

$$= \lim_{y \rightarrow \alpha} \left| \frac{c + by + ay^2}{2y(y - \alpha)} \right|$$

$$= \lim_{y \rightarrow \alpha} \left| \frac{a(y - \alpha)(y - \beta)}{2y(y - \alpha)} \right| = \left| \frac{a(\alpha - \beta)}{2\alpha} \right|$$

$$\text{But } \alpha\beta = \frac{c}{a} \Rightarrow a = \frac{c}{\alpha\beta}$$

$$\text{Hence } \left| \frac{a(\alpha - \beta)}{2\alpha} \right| = \left| \frac{c(\alpha - \beta)}{2\alpha(\alpha\beta)} \right| = \left| \frac{c}{2\alpha} \left(\frac{1}{\alpha} - \frac{1}{\beta} \right) \right|$$

27. If $0 < \alpha < \beta$ then $\lim_{n \rightarrow \infty} (\beta^n + \alpha^n)^{\frac{1}{n}}$ is equal to

(1) α

(2) β

(3) $\alpha\beta$

(4) e

Sol. Answer (2)

$$\lim_{n \rightarrow \infty} \rho \left(1 + \left(\frac{\alpha}{\beta} \right)^n \right)^{\frac{1}{n}}$$

$$= \beta (1 + 0) \quad (\because 0 < \alpha < \beta)$$

$$= \beta$$

28. Let $S_n = \frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots$ to n terms then $\lim_{n \rightarrow \infty} S_n$ is equal to

(1) $\frac{1}{3}$

(2) 3

(3) $\frac{1}{4}$

(4) ∞

Sol. Answer (1)

$$\begin{aligned}\lim_{n \rightarrow \infty} S_n &= \lim_{n \rightarrow \infty} \frac{1}{3} \left[\frac{3}{1.4} + \frac{3}{4.7} + \frac{3}{7.10} + \dots \right] \\&= \lim_{n \rightarrow \infty} \frac{1}{3} \left[\left(1 - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{7} \right) + \left(\frac{1}{7} - \frac{1}{10} \right) + \dots \right] \\&= \lim_{n \rightarrow \infty} \frac{1}{3} \times 1 = \frac{1}{3} \\&= \lim_{n \rightarrow \infty} S_n \\&= \lim_{n \rightarrow \infty} \frac{1}{3} \times 1 \\&= \frac{1}{3}\end{aligned}$$

29. Let $t_r = \frac{r}{1+r^2+r^4}$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n t_r$ equals

(1) $\frac{1}{2}$

(2) 1

(3) $\frac{1}{4}$

(4) $\frac{1}{3}$

Sol. Answer (1)

$$\begin{aligned}t_r &= \frac{r}{1+r^2+r^4} = \frac{r}{r^4+r^2+1+r^2-r^2} = \frac{r}{(r^2+1)^2-r^2} \\t_r &= \frac{2r}{2(r^2+r+1)(r^2-r+1)} \\t_r &= \frac{1}{2} \frac{(r^2+r+1)-(r^2-r+1)}{(r^2+r+1)(r^2-r+1)} \\t_r &= \frac{1}{2} \left[\frac{1}{r^2-r+1} - \frac{1}{r^2+r+1} \right] \\t_1 &= \frac{1}{2} \left[1 - \frac{1}{3} \right] \\t_2 &= \frac{1}{2} \left[\frac{1}{3} - \frac{1}{7} \right]\end{aligned}$$

$$t_3 = \frac{1}{2} \left[\frac{1}{7} - \frac{1}{13} \right]$$

⋮
⋮
⋮

$$t_n = \frac{1}{2} \left[\frac{1}{n^2 - n + 1} - \frac{1}{n^2 + n + 1} \right]$$

$$S_n = \sum_{r=1}^n t_r = t_1 + t_2 + t_3 + \dots + t_n = 1 - \frac{1}{n^2 + n + 1}$$

$$\lim_{n \rightarrow \infty} S_n = \frac{1}{2} \left[1 - \frac{1}{\infty} \right] = \frac{1}{2}$$

30. $\lim_{n \rightarrow \infty} \cos \frac{x}{2} \cos \frac{x}{4} \dots \cos \frac{x}{2^n}$ equals

(1) 1

(2) $\frac{\sin x}{x}$

(3) $\frac{x}{\sin x}$

(4) $\cos x$

Sol. Answer (2)

$$\lim_{n \rightarrow \infty} \cos \frac{x}{2} \cdot \cos \frac{x}{4} \dots \cos \left(\frac{x}{2^n} \right)$$

$$= \lim_{n \rightarrow \infty} \cos \left(\frac{x}{2^n} \right) \cdot \cos \left(\frac{x}{2^{n-1}} \right) \dots \cos \left(\frac{x}{2} \right) \left[\text{using } \cos A \cos 2A \cos 4A \dots \cos (e^{n-1} A) = \frac{\sin(2^n A)}{2^n \sin A} \right]$$

$$= \lim_{n \rightarrow \infty} \left(\frac{\sin \left(\frac{x}{2^n} \cdot 2^n \right)}{2^n \sin \left(\frac{x}{2^n} \right) \cdot \left(\frac{x}{2^n} \right)} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{\sin x}{1 \times x \times 1} = \frac{\sin x}{x}$$

31. $\lim_{n \rightarrow \infty} \left(\tan \theta + \frac{1}{2} \tan \frac{\theta}{2} + \frac{1}{2^2} \tan \frac{\theta}{2^2} + \dots + \frac{1}{2^n} \tan \frac{\theta}{2^n} \right)$ equals

(1) $\frac{1}{\theta}$

(2) $\frac{1}{\theta} - 2 \cot 2\theta$

(3) $2 \cot 2\theta$

(4) $2 \tan^2 \theta$

Sol. Answer (2)

$$\lim_{x \rightarrow \infty} \left(\tan \theta + \frac{1}{2} \tan \frac{\theta}{2} + \frac{1}{2^2} \tan \left(\frac{\theta}{2^2} \right) + \dots + \frac{1}{2^n} \tan \left(\frac{\theta}{2^n} \right) \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2^n} \left(\tan \left(\frac{\theta}{2^n} \right) + 2 \tan \left(\frac{\theta}{2^{n-1}} \right) + \dots + 2^n \tan \theta \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2^n} \left(\cot \left(\frac{\theta}{2^n} \right) - 2^{n+1} \cot 2\theta \right)$$

$$= \lim_{n \rightarrow \infty} \frac{\cot \left(\frac{\theta}{2^n} \right)}{2^n} - \lim_{n \rightarrow \infty} 2 \cot 2\theta$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\left(\tan \left(\frac{\theta}{2^n} \right) \right) \cdot 2^n} \cdot \frac{\theta}{\theta} - 2 \cot 2\theta$$

$$= \lim_{n \rightarrow \infty} \frac{\left(\frac{\theta}{2^n} \right)}{\tan \left(\frac{\theta}{2^n} \right)} \cdot \frac{1}{\theta} - 2 \cot 2\theta = \frac{1}{\theta} - 2 \cot 2\theta$$

32. $\lim_{x \rightarrow 2^+} \left(\frac{[x]^3}{3} - \left[\frac{x}{3} \right]^3 \right)$ is, where $[x]$ represents the integral part of x

(1) 0

(2) $\frac{64}{27}$

(3) $\frac{8}{3}$

(4) $\frac{3}{8}$

Sol. Answer (3)

$$\lim_{x \rightarrow 2^+} \left[\frac{[x]^3}{3} - \left[\frac{x}{3} \right]^3 \right] = \lim_{h \rightarrow 0} \left[\frac{[2+h]^3}{3} - \left[\frac{2-h}{3} \right]^3 \right] = \lim_{h \rightarrow 0} \left[\frac{2^3}{3} - 0 \right] = \frac{8}{3}$$

33. If $\{x\}$ denotes the fractional part of x , then $\lim_{x \rightarrow 0} \frac{\{x\}}{\tan\{x\}}$ is equal to

(1) 1

(2) 0

(3) -1

(4) Limit doesn't exist

Sol. Answer (4)R.H.L. at $(x = 0)$

$$= \lim_{h \rightarrow 0} \frac{\{0+h\}}{\tan\{0+h\}} = \lim_{h \rightarrow 0} \frac{h}{\tan h} = 1$$

L.H.L. at $(x = 0)$

$$= \lim_{h \rightarrow 0} \frac{\{-h\}}{\tan\{-h\}} = \lim_{h \rightarrow 0} \frac{(-h) - [-h]}{\tan((-h) - [-h])} = \lim_{h \rightarrow 0} \frac{1}{\tan 1} = \cot 1$$

Since R.H.L. $\neq$ L.H.L., hence limit does not exist.

34. $\lim_{x \rightarrow 5^+} \frac{x^2 - 9x + 20}{x - [x]}$ is equal to ($[\cdot]$ represent greatest integer function)

- (1) 0 (2) 1 (3) 2 (4) Not define

Sol. Answer (2)

35. If $\lim_{x \rightarrow 0} [1 + x \ln(1 + b^2)]^{1/x} = 2b \sin^2 \theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is

[IIT-JEE 2011]

- (1) $\pm \frac{\pi}{4}$ (2) $\pm \frac{\pi}{3}$ (3) $\pm \frac{\pi}{6}$ (4) $\pm \frac{\pi}{2}$

Sol. Answer (4)

We have,

$$\lim_{x \rightarrow 0} [1 + x \ln(1 + b^2)]^{\frac{1}{x}} = 2b \sin^2 \theta, \quad b > 0 \text{ and } \theta \in (-\pi, \pi)$$

$$\Rightarrow e^{\lim_{x \rightarrow 0} \frac{x \ln(1 + b^2)}{x}} = 2b \sin^2 \theta$$

$$\Rightarrow 1 + b^2 = 2b \sin^2 \theta \leq 2b$$

But $1 + b^2 \geq 2b$, by A.M.-G.M. inequality.

Hence $\sin \theta = \pm 1$

$$\Rightarrow \theta = \pm \frac{\pi}{2}$$

36. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$, then

[IIT-JEE 2012]

- (1) $a = 1, b = 4$ (2) $a = 1, b = -4$ (3) $a = 2, b = -3$ (4) $a = 2, b = 3$

Sol. Answer (2)

$$\text{Here, } \lim_{x \rightarrow \infty} \frac{(x^2 + x + 1) - ax(x + 1) - b(x + 1)}{x + 1} = 4$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{x^2(1 - a) + x(1 - a - b) + 1 - b}{x + 1} = 4$$

This is possible iff

$$1 - a = 0 \Rightarrow a = 1$$

$$\text{and } 1 - a - b = 4 \Rightarrow b = -4$$

37. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation

$$(\sqrt[3]{1+a}-1)x^2 + (\sqrt{1+a}-1)x + (\sqrt[6]{1+a}-1) = 0 \text{ where } a > -1.$$

Then $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are

[IIT-JEE 2012]

(1) $-\frac{5}{2}$ and 1

(2) $-\frac{1}{2}$ and -1

(3) $-\frac{7}{2}$ and 2

(4) $-\frac{9}{2}$ and 3

Sol. Answer (2)

Let $\lim_{a \rightarrow 0^+} \alpha(a) = \alpha; \lim_{a \rightarrow 0^+} \beta(a) = \beta$

$$\text{as, } \alpha + \beta = \lim_{a \rightarrow 0^+} - \left\{ \frac{(1+a)^{\frac{1}{2}} - 1}{(1+a)^{\frac{1}{3}} - 1} \right\} = \frac{-3}{2}$$

$$\text{also, } \alpha\beta = \lim_{a \rightarrow 0^+} \frac{(1+a)^{\frac{1}{6}} - 1}{(1+a)^{\frac{1}{3}} - 1} = \frac{1}{2}$$

Therefore, $\alpha = -\frac{1}{2}, \beta = -1$

SECTION - B

Objective Type Questions (More than one options are correct)

1. Let $f(x) = \frac{(x-1)}{(2x^2-7x+5)}$ then

(1) $\lim_{x \rightarrow 1} f(x) = -\frac{1}{3}$

(2) $\lim_{x \rightarrow 0} f(x) = -\frac{1}{5}$

(3) $\lim_{x \rightarrow \infty} f(x) = 0$

(4) $\lim_{x \rightarrow \frac{5}{2}} f(x)$ does not exist

Sol. Answer (1, 2, 3, 4)

$$f(x) = \frac{x-1}{2x^2-7x+5} = \frac{(x-1)}{2x^2-2x-5x+5} = \frac{(x-1)}{(2x-5)(x-1)}$$

$$f(x) = \frac{(x-1)}{(2x-5)(x-1)} = \frac{1}{2x-5} \text{ if } x \neq 1$$

(1) $\lim_{x \rightarrow 1} \frac{1}{2x-5} = -\frac{1}{3}$

(2) $\lim_{x \rightarrow 0} \frac{1}{2x-5} = -\frac{1}{5}$

(3) $\lim_{x \rightarrow \infty} \frac{1}{2x-5} = 0$

$$(4) \lim_{h \rightarrow 0} f\left(\frac{5}{2} + h\right) = \lim_{h \rightarrow 0} \frac{1}{2\left(\frac{5}{2} + h\right) - 5} = \infty$$

$$\lim_{h \rightarrow 0} f\left(\frac{5}{2} - h\right) = \lim_{h \rightarrow 0} \frac{1}{2\left(\frac{5}{2} - h\right) - 5} = -\infty$$

Hence limit at $x = \frac{5}{2}$ does not exist.

2. $\lim_{x \rightarrow 0} (1 + ax)^{\frac{b}{x}} = e^2$, where $a, b \in N$, then the value of (a, b) is

(1) (16, 8)

(2) (8, 4)

(3) (2, 1)

(4) (1, 2)

Sol. Answer (3, 4)

$$\lim_{x \rightarrow 0} (1 + ax)^{\frac{b}{x}} = e^2 \Rightarrow e^{\lim_{x \rightarrow 0} (1+ax-1)\frac{b}{x}} = e^2$$

$$\Rightarrow ab = 2, \text{ and } a, b \in N$$

$$\Rightarrow \text{either } a = 1, b = 2$$

$$\text{or } a = 2, b = 1$$

3. If $\lim_{x \rightarrow 0} (\cos x + a \sin bx)^{\frac{1}{x}} = e^2$, then (a, b) is equal to

(1) (1, 2)

(2) $\left(2, \frac{1}{2}\right)$

(3) $\left(2\sqrt{3}, \frac{1}{\sqrt{3}}\right)$

(4) (4, 2)

Sol. Answer (1, 3)

$$\lim_{x \rightarrow 0} (\cos x + a \sin bx)^{\frac{1}{x}} = e^2$$

$$= e^{\lim_{x \rightarrow 0} \frac{(\cos x + a \sin bx - 1)}{x}} = e^2$$

$$= \lim_{x \rightarrow 0} \frac{\cos x + a \sin bx - 1}{x} = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{-\sin x + ab \cos bx}{1} = 2 \quad (\text{using L-Hospital rule})$$

$$\Rightarrow ab = 2, \text{ options (1) and (3) satisfies this relation.}$$

4. Let $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{x(1 + p \cos x) - q \sin x}{(f(x))^3} = 1$ then

(1) $p = -\frac{5}{2}$

(2) $q = -\frac{3}{2}$

(3) $2(p^2 + q^2) = 17$

(4) $(p^2 - q^2) = 4$

Sol. Answer (1, 2, 3, 4)

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$$

$$= \lim_{x \rightarrow 0} \frac{x(1 + p \cos x) - q \sin x}{(f(x))^3} = 1$$

$$= \lim_{x \rightarrow 0} \frac{x \left(1 + p \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) \right) - q \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)}{x^3 \cdot \left(\frac{f(x)}{x} \right)^3} = 1$$

$$= \lim_{x \rightarrow 0} \frac{x \left(1 + p - \frac{px^2}{2!} + \frac{px^4}{4!} \dots \right) - q \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} \dots \right)}{x^3 \left(\frac{f(x)}{x} \right)^3} = 1$$

$$= \lim_{x \rightarrow 0} \frac{\left(1 + p - \frac{px^2}{2!} + \frac{px^4}{4!} \dots \right) - q \left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} \dots \right)}{x^2} = 1 \quad \left(\because \lim_{x \rightarrow 0} \frac{f(x)}{x} = 1 \right)$$

$$1 + p - q = 0 \quad \dots (i)$$

$$-\frac{p}{2} + \frac{q}{6} = 1 \quad \dots (ii)$$

By (i) and (ii)

$$p = -\frac{5}{2}, q = -\frac{3}{2}$$

$$\Rightarrow 2(p^2 + q^2) = 17 \text{ and } (p^2 - q^2) = 4$$

5. $\frac{f(3+x^2) - f(3-x^2)}{3 \sin^2 x} = g(x)$ where $f'(x) = 3$ for every x , then

(1) $\lim_{x \rightarrow 0^+} g(x) = 2$

(2) $\lim_{x \rightarrow 0^-} g(x) = 2$

(3) $\lim_{x \rightarrow 0} g(x) = 2$

(4) $\lim_{x \rightarrow 0} g(x)$ does not exist

Sol. Answer (1, 2, 3)

$$\begin{aligned}
 (1) \quad \lim_{x \rightarrow 0^+} g(x) &= \lim_{h \rightarrow 0} g(0+h) = \lim_{h \rightarrow 0} \frac{f(3+h^2) - f(3-h^2)}{3 \sin^2 h} \\
 &= \lim_{h \rightarrow 0} \frac{f'(3+h^2)(2h) - f'(3-h^2)(-2h)}{6 \sin h \cos h} \\
 &= \lim_{h \rightarrow 0} \frac{f'(3+h^2) - f'(3-h^2)}{3 \left(\frac{\sin h}{h} \right) \cdot \cos h} = \frac{f'(3) + f'(3)}{3} = \frac{6}{3} = 2
 \end{aligned}$$

$$(2) \quad \lim_{x \rightarrow 0^-} g(x) = \lim_{h \rightarrow 0} g(0-h) = \lim_{h \rightarrow 0} \frac{f(3+h^2) - f(3-h^2)}{3 \sin^2 h} = 2$$

$$(3) \quad \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^+} g(x) = 2$$

6. Let $f(x) = [x]$ and $[]$ represents the greatest integer function, then

$$(1) \quad \lim_{x \rightarrow 1^+} f(x) = 1$$

$$(2) \quad \lim_{x \rightarrow 2008^-} f(x) = 2007$$

$$(3) \quad \lim_{x \rightarrow K} f(x) = 2008; \quad K \in (2008, 2009)$$

$$(4) \quad \lim_{x \rightarrow 2008^-} x - [x] = 1$$

Sol. Answer (1, 2, 3, 4)

$$(1) \quad \lim_{x \rightarrow 1^+} f(x) = \lim_{h \rightarrow 0} [1+h] = 1$$

$$(2) \quad \lim_{x \rightarrow 2008^-} f(x) = \lim_{h \rightarrow 0} [2008-h] = 2007$$

$$(3) \quad \lim_{x \rightarrow k} f(x) = \lim_{h \rightarrow 0} [k \pm h] = 2008, \text{ as } k \in (2008, 2009)$$

$$(4) \quad \lim_{x \rightarrow 2008^-} (x - [x]) = \lim_{h \rightarrow 0} \{(2008-h) - [2008-h]\} = 2008 - 2007 = 1$$

7. Let $f(x) = \frac{[x] - [\sin x]}{1 - \sin[\cos x]}$ and $g(x) = \frac{1}{f(x)}$ and $[]$ represents the greatest integer function, then

$$(1) \quad \lim_{x \rightarrow 0} f(x) \text{ exists}$$

$$(2) \quad \lim_{x \rightarrow 0} \frac{1}{g(x)} \text{ exists}$$

$$(3) \quad \lim_{x \rightarrow 0} g(x) \text{ does not exist}$$

$$(4) \quad \lim_{x \rightarrow 0} f(x) = 0$$

Sol. Answer (1, 2, 3, 4)

$$f(x) = \frac{[x] - [\sin x]}{1 - \sin[\cos x]}$$

$$(1) \quad \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{[0+h] - [\sin(0+h)]}{1 - \sin[\cos(0+h)]} = \lim_{h \rightarrow 0} \frac{0-0}{1-0} = 0$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{[0-h] - [\sin(0-h)]}{1 - \sin[\cos(0-h)]} = \lim_{h \rightarrow 0} \frac{-1 - (-1)}{1-0} = 0$$

Hence $\lim_{x \rightarrow 0} f(x)$ exist.

$$(2) \lim_{x \rightarrow 0} \frac{1}{g(x)} = \lim_{x \rightarrow 0} f(x), \text{ hence limits exist.}$$

$$(3) \lim_{x \rightarrow 0^+} g(x) = \lim_{h \rightarrow 0} g(0+h) = \lim_{h \rightarrow 0} \frac{1 - \sin[\cos(0+h)]}{[0+h] - [\sin(0+h)]} = \frac{1-0}{0-0} = \infty$$

Hence $\lim_{x \rightarrow 0} g(x)$ does not exist, whatever may be the value of left hand limit.

$$(4) \lim_{x \rightarrow 0} f(x) = 0$$

8. If $y = \tan x \tan 2x \tan 3x$, then $\frac{dy}{dx}$ equals

$$(1) 3\sec^2 3x \tan x \tan 2x + 2\sec^2 2x \tan 3x \tan x + \sec^2 x \tan 2x \tan 3x$$

$$(2) 2y(\operatorname{cosec} 2x + 2 \operatorname{cosec} 4x + 3 \operatorname{cosec} 6x)$$

$$(3) 3\sec^2 3x - 2\sec^2 2x - \sec^2 x$$

$$(4) \sec^2 x + 2\sec^2 2x + 3\sec^2 3x$$

Sol. Answer (1, 2, 3)

$$y = \tan x \cdot \tan 2x \cdot \tan 3x$$

$$\frac{dy}{dx} = \sec^2 x \tan 2x \tan 3x + (\tan x)(2\sec^2 2x)(\tan 3x) + 3(\sec^2 3x)(\tan x)(\tan 2x)$$

$$= \tan x \tan 2x \tan 3x \left[\frac{\sec^2 x}{\tan x} + \frac{\sec^2 2x}{\tan 2x} + \frac{\sec^2 3x}{\tan 3x} \right]$$

$$= y[2 \operatorname{cosec} 2x + 4 \operatorname{cosec} 4x + 6 \operatorname{cosec} 6x]$$

$$= 2y[\operatorname{cosec} 2x + 2 \operatorname{cosec} 4x + 3 \operatorname{cosec} 6x]$$

$$\text{Also } \tan 3x = \frac{\tan x + \tan 2x}{1 - \tan x \tan 2x}$$

$$\Rightarrow \tan 3x - \tan x \cdot \tan 2x \tan 3x = \tan x + \tan 2x$$

$$\Rightarrow \tan x \tan 2x \tan 3x = \tan 3x - \tan 2x - \tan x$$

$$\frac{d}{dx}(\tan x \tan 2x \tan 3x) = 3\sec^2 3x - 2\sec^2 2x - \sec^2 x$$

9. Which of the following statement(s) is/are true?

$$(1) \lim_{x \rightarrow a} \frac{a^x - x^a}{x^x - a^a} = -1 \text{ then } a \text{ is } 1$$

$$(2) \lim_{n \rightarrow \infty} \log_{(n-1)} n \cdot \log_n(n+1) \cdot \log_{(n+1)}(n+2) \cdots \log_{(n^5-1)} n^5 = 5$$

$$(3) \lim_{x \rightarrow 0} \log_{\frac{x^2+1}{x+1}} \left(\sqrt{x^2+1} - x \right) = 1$$

$$(4) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2}{y} = 0, \text{ when } (x, y) \rightarrow (0, 0) \text{ along the curve } y^2 = x^3$$

Sol. Answer (1, 2, 3, 4)

$$(1) \lim_{x \rightarrow a} \frac{a^x - x^a}{x^x - a^a} = \lim_{x \rightarrow a} \frac{a^x \log a - ax^{a-1}}{x^x(1+\log x)} = \frac{a^a \log a - a^a}{a^a(1+\log a)} = \frac{\log a - 1}{1+\log a}$$

$$\text{If } \frac{\log a - 1}{1+\log a} = -1$$

$$\Rightarrow \log a - 1 = -1 - \log a$$

$$2\log a = 0 \Rightarrow a = 1$$

$$(2) \lim_{n \rightarrow \infty} \log_{(n-1)} n \cdot \log_n(n+1) \cdot \log_{(n+1)}(n+2) \dots \log_{(n^5-1)} n^5$$

$$= \lim_{n \rightarrow \infty} \frac{\log n}{\log(n-1)} \cdot \frac{\log(n+1)}{\log n} \cdot \frac{\log(n+2)}{\log(n+1)} \dots \frac{\log n^5}{\log(n^5-1)}$$

$$\lim_{n \rightarrow \infty} \frac{5 \log n}{\log(n-1)} = 5$$

$$(3) \lim_{x \rightarrow 0} \log_{\left(\frac{x^2+1}{x+1}\right)} (\sqrt{x^2+1} - x)$$

$$= \lim_{x \rightarrow 0} \frac{\log(\sqrt{x^2+1} - x)}{\log\left(\frac{x^2+1}{x+1}\right)}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{\sqrt{x^2+1}-x} \cdot \left(\frac{x}{\sqrt{x^2+1}} - 1\right)}{\left(\frac{2x}{x^2+1} - \frac{1}{x+1}\right)}$$

$$\lim_{x \rightarrow 0} \left(\frac{\left(\frac{1}{\sqrt{x^2+1}-x}, \frac{x-\sqrt{1+x^2}}{\sqrt{1+x^2}} \right)}{\frac{2x}{x^2+1} - \frac{1}{x+1}} \right) = \frac{1}{-1} = 1$$

$$(4) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2}{y} = 0, \text{ when } (x, y) \rightarrow (0, 0)$$

$$\lim_{x \rightarrow 0} \frac{x^2}{\sqrt{x^3}} = \lim_{x \rightarrow 0} \frac{x^2}{x^{3/2}} = \lim_{x \rightarrow 0} \sqrt{x} = 0$$

10. The real valued function $f_1(x)$, $f_2(x)$, $f_3(x)$ and $f_4(x)$ are defined as

$$f_1(x) = \begin{cases} x^2, & 0 \leq x < 1 \\ 1, & x \geq 1 \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad \begin{cases} f_2(x) = f_1(-x) \\ f_3(x) = f_2(-x) \\ f_4(x) = f_3(-x) \end{cases}$$

Then

$$(1) \quad f_1(x)f_2(x) = 0 \quad \forall x \in R$$

$$(2) \quad f_2(x)f_3(x) = 0 \quad \forall x \in R$$

$$(3) \quad f_3(x)f_4(x) = 0 \quad \forall x \in R$$

$$(4) \quad f_1(x)f_2(x) = f_3(x)f_4(x)$$

Sol. Answer (1, 2, 3, 4)

$$f_1(x) = \begin{cases} x^2, & 0 < x < 1 \\ 1, & x \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$f_1(-x) = \begin{cases} x^2, & 0 < -x < 1 \\ 1, & -x \geq 1 \\ 0, & \text{otherwise} \end{cases} = \begin{cases} x^2, & -1 < x < 0 \\ 1, & x \leq -1 \\ 0, & \text{otherwise} \end{cases}$$

Hence three cases arises

(i) If $x \leq -1$

$$\text{Then } f_1(x) = 0, \quad f_1(-x) = -1$$

$$\Rightarrow f_1(x)f_1(-x) = 0$$

$$\Rightarrow f_1(x)f_2(x) = 0$$

(ii) If $-1 \leq x \leq 0$

$$f_1(x) = 0$$

$$f_2(x) = f_1(-x) = x^2$$

$$\Rightarrow f_1(x)f_2(x) = 0$$

(iii) If $0 \leq x \leq 1$

$$f_1(x) = x^2$$

$$f_1(-x) = 0$$

$$\Rightarrow f_1(x)f_2(x) = 0$$

The same condition may be applied for $f_2(-x)$ and $f_3(-x)$

$$\text{Hence } f_2(x)f_2(-x) = 0 \Rightarrow f_2(x)f_3(x) = 0$$

$$\text{And } f_3(x)f_3(-x) = 0 \Rightarrow f_3(x)f_4(x) = 0$$

$$\text{or } f_1(x)f_2(x) = f_3(x)f_4(x)$$

11. If $f(n+1) = \frac{1}{2} \left\{ f(n^2 - 1) + \frac{9}{f(n)} \right\}$, $n \in \mathbb{N}$ and $f(n) > 0$ for all $n \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} f(n)$ will be equal to if it exists

(1) 3

(2) -3

(3) $\frac{1}{2}$

(4) None of these

Sol. Answer (1, 2)

$$\text{If } \lim_{x \rightarrow \infty} f(x) = L$$

$$\text{Then } \lim_{x \rightarrow \infty} f(x^2 - 1) = L$$

$$\lim_{x \rightarrow \infty} f(x+1) = L$$

$$\text{Here } L = \frac{1}{2} \left(L + \frac{9}{L} \right)$$

$$\frac{1L}{2} = \frac{9}{2L}$$

$$L^2 = 9$$

$$L = \pm 3$$

12. $\lim_{x \rightarrow \infty} x^2 \sin \left(\log \sqrt{\cos \frac{\pi}{x}} \right) =$

(1) $\frac{\pi^2}{4}$ (2) $-\frac{\pi^2}{4}$ (3) $\frac{\pi^2}{8}$ (4) $-\frac{\pi^2}{8}$

Sol. Answer (2)

$$\lim_{x \rightarrow \infty} \frac{x^2 \sin \left(\log \sqrt{\cos \frac{\pi}{x}} \right)}{\log \sqrt{\cos \frac{\pi}{x}}} \log \sqrt{\cos \frac{\pi}{x}}$$

$$= \lim_{x \rightarrow \infty} x^2 \log \sqrt{\cos \frac{\pi}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{\log \sqrt{\cos \frac{\pi}{x}}}{\frac{1}{x^2}} \quad \frac{0}{0} \text{ form}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{\sqrt{\cos \frac{\pi}{x}}} \cdot \frac{1}{2} \left(\cos \frac{\pi}{x} \right)^{-\frac{1}{2}} \sin \frac{\pi}{x} \left(-\frac{\pi}{x^2} \right)}{-\frac{2}{x^3}}$$

$$= -\frac{\pi^2}{4}$$

13. $\lim_{x \rightarrow 0} \frac{1}{x^2} \int_0^{x^2} \frac{\cos x \cos \sqrt{\theta}}{1 + \sin^2 \sqrt{\theta}} d\theta$ equals to

(1) 0

(2) 1

(3) -1

(4) Does not exist

Sol. Answer (2)

$$\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \frac{\cos x \cos \sqrt{\theta}}{1 + \sin^2 \sqrt{\theta}} d\theta}{x^2} \quad \frac{0}{0} \text{ form}$$

$$\lim_{x \rightarrow 0} \frac{\frac{\cos^2 x}{1 + \sin^2 x} \cdot 2x}{2x} = 1$$

14. If $f(n)$ is continuous in $[0, 1]$ and $f\left(\frac{1}{2}\right) = 1$ then $\lim_{n \rightarrow \infty} f\left(n - \sqrt{n^2 - n}\right)$ is

(1) 1

(2) $\frac{1}{2}$

(3) -1

(4) Cannot be determine

Sol. Answer (1)

$$\begin{aligned} \lim_{n \rightarrow \infty} f(n - \sqrt{n^2 - n}) \\ &= f\left(\lim_{n \rightarrow \infty} (n - \sqrt{n^2 - n})\right) \\ &= f\left(\lim_{n \rightarrow \infty} \frac{n^2 - n^2 + n}{n + \sqrt{n^2 - n}}\right) \\ &= f\left(\frac{1}{2}\right) = 1 \end{aligned}$$

15. Let $f: \mathbb{R} \rightarrow (0, \infty)$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable functions such that f'' and g'' are continuous functions on $\mathbb{R}$. Suppose $f'(2) = g(2) = 0$, $f''(2) \neq 0$ and $g'(2) \neq 0$. If $\lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1$, then

[JEE(Advanced)-2016](1) f has a local minimum at $x = 2$ (2) f has a local maximum at $x = 2$ (3) $f''(2) > f(2)$ (4) $f(x) - f''(x) = 0$ for at least one $x \in \mathbb{R}$ **Sol.** Answer (1, 4)

$$f'(2) = g(2) = 0 \quad f''(2) \neq 0$$

$$\text{Given, } \lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1$$

$$\frac{0}{0} \text{ form we can apply L' Hospital rule}$$

$$\lim_{x \rightarrow 2} \frac{f(x)g'(x) + f'(x) \cdot g(x)}{f'(x)g''(x) + g'(x) \cdot f''(x)}$$

$$\lim_{x \rightarrow 2} \frac{f(2)g'(2) + f'(2) \cdot g(2)}{f'(2)g''(2) + g'(2) \cdot f''(2)} = 1$$

$$\lim_{x \rightarrow 2} \frac{f(2)g'(2)}{g'(2) \cdot f''(2)} = 1$$

$$f(2) = f''(2)$$

Option (4) is correct.

$f''(2)$ = positive from question range is positive

It is local minimum option (1) is correct.

16. Let $f(x) = \lim_{n \rightarrow \infty} \left[\frac{n^n (x+n)(x+\frac{n}{2}) \dots (x+\frac{n}{n})}{n!(x^2+n^2)(x^2+\frac{n^2}{4}) \dots (x^2+\frac{n^2}{n^2})} \right]^{\frac{x}{n}}$, for all $x > 0$. Then

[JEE(Advanced)-2016]

(1) $f\left(\frac{1}{2}\right) \geq f(1)$

(2) $f\left(\frac{1}{3}\right) \leq f\left(\frac{2}{3}\right)$

(3) $f'(2) \leq 0$

(4) $\frac{f'(3)}{f(3)} \geq \frac{f'(2)}{f(2)}$

Sol. Answer (2, 3)

$$f(x) = \lim_{n \rightarrow \infty} \left[\frac{n^n \cdot (x+n) \left(x+\frac{n}{2}\right) \left(x+\frac{n}{3}\right) \dots \left(x+\frac{n}{n}\right)}{n!(x^2+n^2) \left(x^2+\frac{n^2}{4}\right) \dots \left(x^2+\frac{n^2}{n^2}\right)} \right]^{\frac{x}{n}}$$

$$f(x) = \lim_{n \rightarrow \infty} \left[\frac{\left(x + \frac{1}{\left(\frac{1}{n}\right)}\right) \left(x + \frac{1}{\left(\frac{2}{n}\right)}\right) \left(x + \frac{1}{\left(\frac{3}{n}\right)}\right) \dots \left(x + \frac{1}{\left(\frac{n}{n}\right)}\right)}{\left(\frac{1}{n} \cdot \frac{2}{n} \cdot \frac{3}{n} \dots \frac{n}{n}\right) \left(x^2 + \frac{1}{\left(\frac{1}{n^2}\right)}\right) \left(x^2 + \frac{1}{\left(\frac{4}{n^2}\right)}\right) \dots \left(x^2 + \frac{1}{\left(\frac{n^2}{n^2}\right)}\right)} \right]^{\frac{x}{n}}$$

$$\log f(x) = \lim_{n \rightarrow \infty} \frac{x}{n} \left[\sum_{r=1}^n \left(\log \left(x + \frac{1}{\frac{r}{n}} \right) - \log \left(x^2 + \frac{1}{\frac{r^2}{n^2}} \right) - \log \frac{r}{n} \right) \right]$$

$$\begin{aligned}
 \log f(x) &= x \left[\int_0^1 \log \left(x + \frac{1}{t} \right) dt - \int_0^1 \log \left(x^2 + \frac{1}{t^2} \right) dt - \int_0^1 \log t dt \right] \\
 &= x \int_0^1 \log \left(\frac{\left(x + \frac{1}{t} \right)}{\left(x^2 + \frac{1}{t^2} \right) \cdot t} \right) dt \\
 &= x \int_0^1 \log \left(\frac{tx + 1}{t^2 x^2 + 1} \right) dt
 \end{aligned}$$

Put $tx = z \Rightarrow xdt = dz$

$$\log f(x) = \int_0^x \log \left(\frac{1+z}{1+z^2} \right) dz$$

$$\frac{f'(x)}{f(x)} = \log \left(\frac{1+x}{1+x^2} \right)$$

for $x \in (0, 1)$, $\ln(1+x) > \ln(1+x^2)$

and for $x \in (1, \infty)$ $\ln(1+x) < \ln(1+x^2)$

So, $f'(x) > 0$ for $x \in (0, 1) \Rightarrow f(x)$ is increasing

and $f'(x) < 0$ for $x \in (1, \infty) \Rightarrow f(x)$ is decreasing

$$\text{So, } f\left(\frac{1}{2}\right) \leq f(1), f\left(\frac{1}{3}\right) \leq f\left(\frac{2}{3}\right), f'(2) < 0$$

$$\frac{f'(3)}{f(3)} = \ln\left(\frac{2}{5}\right) \text{ and } \frac{f'(2)}{f(2)} = \ln\left(\frac{3}{5}\right)$$

$$\text{So, } \frac{f'(2)}{f(2)} > \frac{f'(3)}{f(3)}$$

17. Let $[x]$ be the greatest integer less than or equals to x . Then, at which of the following point(s) the function $f(x) = x \cos(\pi(x + [x]))$ is discontinuous? [JEE(Advanced)-2017]

- (1) $x = 1$ (2) $x = -1$ (3) $x = 0$ (4) $x = 2$

Sol. Answer (1, 2, 4)

$$f(x) = x \cos(\pi(x + [x]))$$

$$= \begin{cases} x \cos \pi x, [x] \text{ is even} \\ -x \cos \pi x, [x] \text{ is odd} \end{cases}$$

Clearly $f(1^+) \neq f(1)$, $f(2^+) \neq f(2)$, $f(-1^+) \neq f(-1^-)$

but $f(0) = f(0^+) = f(0^-) = 0$ hence f is discontinuous at $x = 1, -1, 2$ but continuous at $x = 0$.

18. Let $f(x) = \frac{1-x(1+|1-x|)}{|1-x|} \cos\left(\frac{1}{1-x}\right)$ for $x \neq 1$. Then

[JEE(Advanced)-2017]

(1) $\lim_{x \rightarrow 1^+} f(x) = 0$

(2) $\lim_{x \rightarrow 1^-} f(x) = 0$

(3) $\lim_{x \rightarrow 1^+} f(x)$ does not exist

(4) $\lim_{x \rightarrow 1^-} f(x)$ does not exist

Sol. Answer (2, 3)

$$f(x) = \frac{1-x(1+|1-x|)}{|1-x|} \cos\left(\frac{1}{1-x}\right)$$

$$\lim_{x \rightarrow 1^+} \frac{1-x(1+x-1)}{(x-1)} \cos\left(\frac{1}{1-x}\right)$$

$$\lim_{x \rightarrow 1^+} \frac{1-x^2}{(x-1)} \cos\left(\frac{1}{1-x}\right)$$

$$\lim_{x \rightarrow 1^+} -(1+x) \cos\left(\frac{1}{1-x}\right) = \text{a number lying between } -2 \text{ and } 2$$

Hence, limit does not exist.

$$\lim_{x \rightarrow 1^-} \frac{1-x(1+(1-x))}{(1-x)} \cos\left(\frac{1}{1-x}\right)$$

$$\lim_{x \rightarrow 1^-} \frac{1-x(2-x)}{(1-x)} \cos\left(\frac{1}{1-x}\right)$$

$$\lim_{x \rightarrow 1^-} (1-x) \cos\left(\frac{1}{1-x}\right) = 0$$

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

A square is inscribed in a circle of radius R , a circle is inscribed in the square, a new square in the circle and so on for n times.

1. Sum of the areas of all circles is

(1) $4\pi R^2 \left(1 - \left(\frac{1}{2}\right)^n\right)$

(2) $2\pi R^2 \left(1 - \left(\frac{1}{2}\right)^n\right)$

(3) $3\pi R^2 \left(1 - \left(\frac{1}{3}\right)^n\right)$

(4) $\pi R^2 \left(1 - \left(\frac{1}{2}\right)^n\right)$

2. The limit of sum of areas of all the squares as $n \rightarrow \infty$ is

(1) $2R^2$

(2) $3R^2$

(3) $4R^2$

(4) $8R^2$

3. The limit of sum of areas of all the circles as $n \rightarrow \infty$

(1) $2\pi R^2$

(2) $3\pi R^2$

(3) $4\pi R^2$

(4) $8\pi R^2$

Solution of Comprehension-I

$$\text{Radius of circle } C_1 = R \Rightarrow \text{Area } (A_1) = \pi R^2$$

$$\text{Sides of the squares } S_1 = \frac{2R}{\sqrt{2}} = R\sqrt{2}$$

$$\text{Radius of circle } C_2 = \frac{R\sqrt{2}}{2} = \frac{R}{\sqrt{2}}$$

$$\Rightarrow A_2 = \frac{\pi R^2}{2}$$

$$\text{Similarly } A_3 = \frac{\pi R^2}{4}, A_4 = \frac{\pi R^2}{8}, \dots$$

$$A = \text{Sum of area} = A_1 + A_2 + \dots + A_n$$

$$= \pi R^2 \left[1 + \frac{1}{2} + \frac{1}{4} + \dots \text{upto } n \text{ terms} \right]$$

$$= \pi R^2 \left[\frac{1 - \left(\frac{1}{2}\right)^n}{1 - \frac{1}{2}} \right] = 2\pi R^2 \left[1 - \left(\frac{1}{2}\right)^n \right]$$

$$\text{Let area of squares be } B_1, B_2, \dots, B_n$$

$$\Rightarrow B_1 = (R\sqrt{2})^2 = 2R^2$$

$$B_2 = R^2$$

$$B_3 = \frac{R^2}{2} \dots \text{and so on upto } B_n$$

$$\text{Sum of area of squares} = B_1 + B_2 + B_3 + \dots + B_n$$

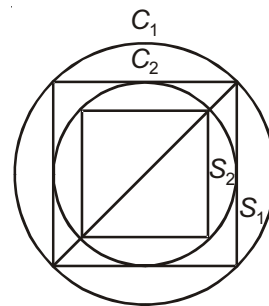
$$B = 2R^2 + R^2 + \frac{R^2}{2} + \dots \text{upto } n \text{ terms}$$

$$= R^2 \left[2 + 1 + \frac{1}{2} + \dots \text{upto } n \text{ terms} \right]$$

$$= R^2 \left[2 \cdot \left(\frac{1}{2}\right)^{n-1} \right] = 2R^2 \left(\frac{1}{2}\right)^{n-1}$$

$$\text{Now, } \lim_{n \rightarrow \infty} B = \lim_{n \rightarrow \infty} 2R^2 \left(\frac{1}{2}\right)^{n-1} = 2R^2 \cdot 0 = 0$$

$$\lim_{n \rightarrow \infty} 2\pi R^2 \left[1 - \left(\frac{1}{2}\right)^n \right] = 2\pi R^2$$



1. Answer (2)
2. Answer (3)
3. Answer (1)

Comprehension-II

The notion of limit of a single variable function can be extended to the function of two variables. When $(x, y) \rightarrow (a, b)$ i.e., $x \rightarrow a$, $y \rightarrow b$ simultaneously, then $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$ can be evaluated by the path along with (x, y) approaches to (a, b) .

For example $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2}{y}$ where $(x, y) \rightarrow (0, 0)$ along the curve $y^2 = x^3$ can be evaluated as

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2}{y} = \lim_{\substack{x \rightarrow 0 \\ y = x^{3/2}}} \frac{x^2}{x^{3/2}} = \lim_{x \rightarrow 0} x^{1/2} = 0$$

Then,

1. $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^3}{y}$ where $(x, y) \rightarrow (0, 0)$ along the curve $y^2 = x^5$ equals

(1) 0

(2) 1

(3) 2

(4) 3

Sol. Answer (1)

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^3}{y} = \lim_{x \rightarrow 0} \frac{x^3}{x^{5/2}} = 0$$

2. $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 + \sin y}{y^2 + \sin x^2}$ when $(x, y) \rightarrow (0, 0)$ along the curve $y = x^2$ is equal to

(1) 0

(2) 1

(3) 2

(4) 3

Sol. Answer (3)

$$\begin{aligned} & \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 + \sin y}{y^2 + \sin x^2} \\ &= \lim_{x \rightarrow 0} \frac{x^2 + \sin x^2}{x^4 + \sin x^2} \\ &= \lim_{x \rightarrow 0} \frac{x^2 + \left(\frac{\sin x^2}{x^2} \right) x^2}{x^4 + \left(\frac{\sin x^2}{x^2} \right) x^2} \\ &= \lim_{x \rightarrow 0} \frac{x^2 + x^2}{x^4 + x^2} \\ &= \lim_{x \rightarrow 0} \frac{2}{x^2 + 1} = 2 \end{aligned}$$

3. $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 0}} \frac{y^3}{x^3 - y^2 - 1}$, when $(x, y) \rightarrow (1, 0)$ along the line $y = x - 1$ is equal to

(1) 0

(2) 1

(3) 2

(4) 3

Sol. Answer (1)

$$\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 0}} \frac{y^3}{x^3 - y^2 - 1} = \lim_{x \rightarrow 1} \frac{(x-1)^3}{x^3 - (x-1)^2 - 1} \lim_{x \rightarrow 1} \frac{3(x-1)^2}{3x^2 - 2(x-1)} = 0$$

Comprehension-III

We have

$$f(x) = \lim_{n \rightarrow \infty} \cos \frac{x}{2} \cos \frac{x}{2^2} \cos \frac{x}{2^3} \cos \frac{x}{2^4} \dots \cos \frac{x}{2^n} = \frac{\sin x}{2^n \sin \frac{x}{2^n}}$$

using the identity

1. $\lim_{n \rightarrow \infty} \lim_{x \rightarrow 0} f(x)$ equals

(1) 0

(2) 1

(3) 2

(4) $\frac{1}{2}$

2. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{2^k} \tan\left(\frac{x}{2^k}\right)$ equals

(1) $\frac{1}{x} - \tan x$ (2) $\frac{1}{x} - \cot x$ (3) $\frac{1}{x} + \cot x$ (4) $\frac{1}{x} + \tan x$

3. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{2^{2k}} \sec^2\left(\frac{x}{2^k}\right)$ equals

(1) $\operatorname{cosec}^2 x - \frac{1}{x^2}$ (2) $\operatorname{cosec}^2 x + \frac{1}{x}$ (3) $\operatorname{cosec}^2 x - \frac{1}{2x}$ (4) $\operatorname{cosec}^2 x + \frac{1}{2x}$ **Solution of Comprehension-III**

$$\cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \dots \cos\left(\frac{x}{2^n}\right) = \frac{\sin x}{2^n \sin\left(\frac{x}{2^n}\right)} = f(x)$$

1. Answer (2)

$$\lim_{n \rightarrow \infty} \lim_{x \rightarrow 0} \frac{\sin x}{2^n \sin\left(\frac{x}{2^n}\right)} = \lim_{n \rightarrow \infty} \frac{(\sin x)}{2^n \cdot \left(\frac{\sin\left(\frac{x}{2^n}\right)}{\left(\frac{x}{2^n}\right)}\right) \cdot \left(\frac{x}{2^n}\right)} = \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

2. Answer (2)

$$\cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \dots \cos \frac{x}{2^n} = \frac{\sin x}{2^n \sin \left(\frac{x}{2^n} \right)}$$

$$\log \left(\cos \frac{x}{2} \right) + \log \left(\cos \frac{x}{2^2} \right) + \dots = \log(\sin x) - \log \left(2^n \cdot \sin \left(\frac{x}{2^n} \right) \right) \quad \text{differentiating the whole equation.}$$

$$-\frac{1}{2} \cdot \tan \left(\frac{x}{2} \right) - \frac{1}{2^2} \tan \left(\frac{x}{2^2} \right) - \dots = \cot x - \frac{1}{2^n} \cot \left(\frac{x}{2^n} \right)$$

$$\Rightarrow \frac{1}{2} \tan \frac{x}{2} + \frac{1}{2^2} \tan \left(\frac{x}{2^2} \right) + \dots = \frac{1}{2^n} \cot \left(\frac{x}{2^n} \right) - \cot x \quad \dots (i)$$

$$\Rightarrow \sum_{k=1}^n \frac{1}{2^k} \tan \left(\frac{x}{2^k} \right) = \frac{1}{2^n} \cot \left(\frac{x}{2^n} \right) - \cot x$$

3. Answer (1)

Again differentiating equation (i) of Q. 2, we get

$$\frac{1}{2^2} \sec^2 \frac{x}{2} + \frac{1}{2^4} \sec^2 \left(\frac{x}{2^2} \right) + \dots = -\frac{1}{2^{2n}} \operatorname{cosec}^2 \left(\frac{x}{2^n} \right) + \operatorname{cosec}^2 x$$

Comprehension-IVThe greatest integer function $[x]$ is defined as the greatest integer $\leq x$ and $\{x\}$ is defined as $\{x\} = x - [x]$.1. $\lim_{x \rightarrow 0^+} \frac{a^{2[x]+\{x\}} - 1}{2[x] + \{x\}}$ is equal to

(1) $\ln a$

(2) $1 - a$

(3) $1 - \frac{1}{a}$

(4) $1 + \frac{1}{a}$

Sol. Answer (1)

$$\lim_{x \rightarrow 0^+} \frac{a^{2[x]+\{x\}} - 1}{2[x] + \{x\}} = \lim_{h \rightarrow 0} \frac{a^{2[0+h]+\{h\}} - 1}{2[0+h] + \{h\}} = \lim_{h \rightarrow 0} \frac{a^h - 1}{h} = \log a$$

2. $\lim_{x \rightarrow 0^-} \frac{a^{2[x]+\{x\}} - 1}{2[x] + \{x\}}$ is equal to

(1) $\ln a$

(2) $1 - a$

(3) $1 - \frac{1}{a}$

(4) $1 + \frac{1}{a}$

Sol. Answer (3)

$$\begin{aligned} \lim_{x \rightarrow 0^-} \frac{a^{2[x]+\{x\}} - 1}{2[x] + \{x\}} \\ = \lim_{h \rightarrow 0} \frac{a^{2[0-h]+\{0-h\}} - 1}{2[0-h] + \{0-h\}} = \lim_{h \rightarrow 0} \frac{a^{-1} - 1}{-1} = 1 - \frac{1}{a} \end{aligned}$$

3. Let $f(x) = [x]^2 - [x^2]$, then

(1) $\lim_{x \rightarrow 2} f(x) = 0$

(2) $\lim_{x \rightarrow 1} f(x) = 0$

(3) $\lim_{x \rightarrow 0} f(x) = 1$

(4) $\lim_{x \rightarrow 1} f(x)$ does not exist

Sol. Answer (2)

$$\lim_{x \rightarrow 1} [x]^2 - [x^2]$$

Here we may generalize the result at $x = k$ (a integer)

$$\text{R.H.L. at } (x = k) = \lim_{h \rightarrow 0} [k + h]^2 - [(k + h)^2] = k^2 - k^2 = 0$$

$$\text{L.H.L. at } (x = k) = \lim_{h \rightarrow 0} [k - h]^2 - [(k - h)^2] = (k - 1)^2 - (k^2 - 1) = 2k - 2.$$

If R.H.L. = L.H.L.

$$\Rightarrow 2k - 2 = 0 \Rightarrow k = 1$$

Hence $\lim_{x \rightarrow 1} f(x)$ exist and $\lim_{x \rightarrow 1} f(x) = 0$

Comprehension-V

Let $f(x) = \min\{x - [x], -x - [-x]\}$, $-2 \leq x \leq 2$

$$g(x) = |2 - |x - 2||, -2 \leq x \leq 2$$

and $h(x) = \frac{|\sin x|}{\sin x}$, $-2 \leq x \leq 2$ and $x \neq 0$. be three given functions where $[x]$ denotes the greatest integer $\leq x$. Then

1. The number of solution(s) of the equation $x^2 + (f(x))^2 = 1$ ($-1 \leq x \leq 1$) is/are

(1) 0

(2) 2

(3) 4

(4) 6

2. The range of $f(x)$ is

(1) $\left[0, \frac{1}{2}\right]$

(2) $[0, 1]$

(3) $[0, 2]$

(4) $\left[-\frac{1}{2}, 0\right]$

3. The sum of all the roots of the equation $g(x) - h(x) = 0$ is

(1) Zero

(2) Positive

(3) Negative

(4) Not possible

Solution of Comprehension-V

$$f(x) = \min\{x - [x], -x - [-x]\}, -2 \leq x \leq 2$$

$$g(x) = |2 - |x - 2||, -2 \leq x \leq 2$$

$$= |2 - (2 - x)| = |x|$$

$$h(x) = \frac{|\sin x|}{\sin x}, -2 \leq x \leq 2$$

$$h(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 2 \\ -1 & \text{if } -2 \leq x \leq 0 \end{cases}$$

1. Answer (2)

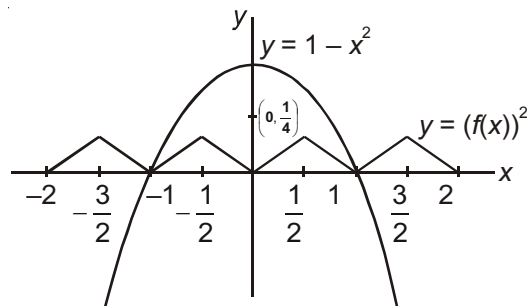
$$x^2 + (f(x))^2 = 1$$

$$(f(x))^2 = 1 - x^2$$

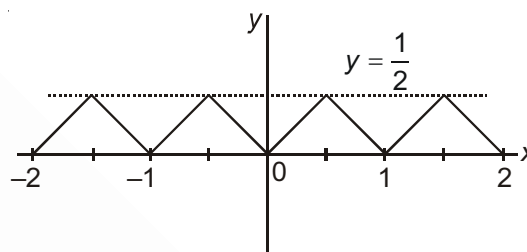
To find the number of solution we draw two graphs

$$y = (f(x))^2 \text{ and } y = 1 - x^2$$

$$(i) \quad f(x) = \min\{\{x\}, \{-x\}\}$$

The graph of $y = (f(x))^2$ and $y = 1 - x^2$ can be shown asHence only two solutions $x = \pm 1$ exists.

2. Answer (1)

The graph of $y = f(x)$ is given byClearly the range of $y = f(x)$ is $\left[0, \frac{1}{2}\right]$.

3. Answer (2)

Comprehension-VILet $\lim_{x \rightarrow 0} f(x)$ be a finite number, where

$$f(x) = \frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}, a, b, c \in \mathbb{R}$$

1. The value of a is

(1) $-\frac{1}{2}$

(2) 0

(3) $\frac{1}{2}$

(4) 1

2. The value of b is

(1) $-\frac{1}{2}$

(2) $\frac{1}{2}$

(3) 0

(4) 1

3. The value of c is

(1) $-\frac{1}{2}$

(2) $\frac{1}{2}$

(3) 0

(4) 2

Solution of Comprehension-VI

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} \dots\right) + a\left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots\right) + b\left(1 - \frac{x}{1!} + \frac{x^2}{2!} - \dots\right) + c\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots\right)}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{(a+b) + x(1+a-b) + x^2\left(\frac{a}{2} + \frac{b}{2} - \frac{c}{2}\right) + x^3\left(-\frac{1}{6} + \frac{a}{6} - \frac{b}{6}\right) + x^4\left(\frac{a}{24} + \frac{b}{24} + \frac{c}{4}\right) + \dots}{x^3}$$

If $\lim_{x \rightarrow 0} f(x)$ infinite then

$$a + b = 0 \quad \dots (i)$$

$$1 + a - b = 0 \quad \dots (ii)$$

$$\frac{a}{2} + \frac{b}{2} - \frac{c}{2} = 0 \quad \dots (iii)$$

By (i), (ii) and (iii),

$$a = -\frac{1}{2}$$

$$b = \frac{1}{2}$$

$$c = 0$$

1. Answer (1)

2. Answer (2)

3. Answer (3)

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x}$ does not exist.

and

STATEMENT-2 : For $\frac{\sqrt{1 - \cos 2x}}{x}$ at $(x = 0)$. Right hand limit $\neq$ Left hand limit

Sol. Answer (1)

$$f(x) = \frac{\sqrt{1 - \cos 2x}}{x} = \sqrt{2} \frac{|\sin x|}{x}$$

$$f(0^+) = \lim_{h \rightarrow 0} \sqrt{2} \frac{|\sin h|}{h} = \sqrt{2}$$

$$f(0^-) = \lim_{h \rightarrow 0} \frac{\sqrt{2} |\sin(0 - h)|}{(-h)} = -\sqrt{2}$$

$f(0^+) \neq f(0^-)$, hence at $x = 0$, limit does not exist.

2. STATEMENT-1 : $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x} = \sqrt{2}$.

and

$$\text{STATEMENT-2 : } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

Sol. Answer (4)

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x} = \lim_{x \rightarrow 0} \frac{\sqrt{2} |\sin x|}{x}$$

$$\text{R.H.L. at } (x = 0) = \lim_{h \rightarrow 0} \frac{\sqrt{2} |\sin(0 + h)|}{h} = \sqrt{2}$$

$$\text{L.H.L. at } (x = 0) = \lim_{h \rightarrow 0} \frac{\sqrt{2} |\sin(0 - h)|}{(0 - h)} = -\sqrt{2}$$

Hence limit does not exist.

Statement-1 is false. But statement-2 is clearly true.

3. STATEMENT-1 : $\lim_{x \rightarrow 0} \frac{\tan x^\circ}{x} = 1$, where x° means x degrees.

and

STATEMENT-2 : If $\lim_{x \rightarrow 0} f(x) = l$, $\lim_{x \rightarrow 0} g(x) = m$, then $\lim_{x \rightarrow 0} \{f(x)g(x)\} = lm$.

Sol. Answer (4)

$$\lim_{x \rightarrow 0} \frac{\tan x^\circ}{x} = \lim_{x \rightarrow 0} \frac{\tan\left(\frac{\pi x}{180}\right)}{x} = \frac{\pi}{180}$$

Statement-1 is false and statement-2 is true.

4. STATEMENT-1 : $\lim_{x \rightarrow 0} [x]$ does not exist, where $[x]$ represent the greatest integer function.

and

STATEMENT-2 : $\lim_{x \rightarrow 0^+} \frac{x[x]}{\sin |x|} = 0$, where $[x]$ denotes the greatest integer function.

Sol. Answer (2)

$$f(x) = [x]$$

$$f(0^+) = \lim_{h \rightarrow 0} [0 + h] = 0$$

$$\text{RHL} = f(0^-) = \lim_{h \rightarrow 0} [0 - h] = -1$$

$$\text{LHL} = \text{Hence } \lim_{x \rightarrow 0} [x] \text{ does not exist.}$$

Statement-1 is true.

$$\lim_{x \rightarrow 0^+} \frac{x[x]}{\sin |x|} = \lim_{h \rightarrow 0} \frac{(0 + h)[0 + h]}{\sin |0 + h|} = \lim_{h \rightarrow 0} \frac{(0 + h) \times 0}{\sin |0 + h|} = 0$$

Hence statement-2 is true.

5. STATEMENT-1 : $\lim_{x \rightarrow 0^-} \frac{\sin[x]}{[x]} = \sin 1, [x] \neq 0$, where $[x]$ is the integral part of x .

and

STATEMENT-2 : $\lim_{x \rightarrow 0^+} \frac{\sin[x]}{[x]} \neq 1, [x] \neq 0$, where $[x]$ is the integral part of x .

Sol. Answer (2)

$$\lim_{x \rightarrow 0^-} \frac{\sin[x]}{[x]} = \lim_{h \rightarrow 0} \frac{\sin[0-h]}{[0-h]} = \lim_{h \rightarrow 0} \frac{-\sin 1}{-1} = \sin 1$$

Statement-1 is true.

$$\lim_{x \rightarrow 0^+} \frac{\sin[x]}{[x]} = \lim_{h \rightarrow 0} \frac{\sin[0+h]}{[0+h]} = \frac{0}{0} \neq 1$$

Statement-2 is true.

6. STATEMENT-1 : $\lim_{x \rightarrow \infty} \left(\frac{x-1}{x+1} \right)^{x+2} = e^{-2}$.

and

STATEMENT-2 : $\lim_{x \rightarrow \infty} \frac{x^5}{(2008)^x} = 120$.

Sol. Answer (3)

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(\frac{x-1}{x+1} \right)^{x+2} &= e^{\lim_{x \rightarrow \infty} \left(\frac{x-1}{x+1} - 1 \right)(x+2)} \\ &= e^{\lim_{x \rightarrow \infty} \left(-\frac{2}{x+1} \right)(x+2)} \\ &= e^{-2} \end{aligned}$$

Statement-1 is true.

$$\lim_{x \rightarrow \infty} \frac{x^5}{(2008)^x} = \lim_{x \rightarrow \infty} \frac{120}{(2008)^x (\log 2008)^5} = 0$$

Hence statement-2 is false.

7. STATEMENT-1 : $\lim_{x \rightarrow \infty} \left(1 + \sin \frac{p}{x} \right)^x = e^p$.

and

STATEMENT-2 : $\lim_{x \rightarrow \infty} \left(1 + \frac{p}{x} \right)^x = e^p$.

Sol. Answer (1)

$$\lim_{x \rightarrow \infty} \left(1 + \sin \frac{p}{x}\right)^x = e^{\lim_{x \rightarrow \infty} \left(1 + \sin \frac{p}{x} - 1\right)x}$$

$$= e^{\lim_{x \rightarrow \infty} \frac{\sin\left(\frac{p}{x}\right)}{\left(\frac{p}{x}\right)} \left(\frac{p}{x}\right)x}$$

$$= e^p$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{p}{x}\right)^x = e^{\lim_{x \rightarrow \infty} \left(1 + \frac{p}{x} - 1\right)x}$$

$$= e^p$$

8. STATEMENT-1 : $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x(2^x - 1)} = \frac{1}{2} \log_2 e$.

and

STATEMENT-2 : $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_a a$, $a > 0$

Sol. Answer (1)

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x(2^x - 1)} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x \left(\frac{2^x - 1}{x}\right) \cdot x}$$

$$= \lim_{x \rightarrow 0} \frac{2 \cdot \left(\frac{\sin\left(\frac{x}{2}\right)}{\frac{x}{2}}\right)^2 \cdot \frac{x^2}{4}}{x \left(\frac{2^x - 1}{x}\right)^2 \cdot x}$$

$$= \lim_{x \rightarrow 0} \frac{2 \times \frac{1}{4} \times \left(\frac{\sin\left(\frac{x}{2}\right)}{\frac{x}{2}}\right)^2}{\left(\frac{2^x - 1}{x}\right)^2}$$

$$= \frac{1}{2} \frac{1}{\log_e 2} = \frac{1}{2} \log_2 e$$

Statement-2 is clearly true and correct explanation of statement-1.

SECTION - E

Matrix-Match Type Questions

1. Match the limit in column I with their values in column II.

Column-I

Column-II

(A) $\lim_{x \rightarrow 0} \frac{\cos x - \cos 3x}{x(\sin x - \sin 3x)}$

(p) 3

(B) $\lim_{x \rightarrow 0} \frac{\sin 4x}{\tan 2x}$

(q) -2

(C) $\lim_{x \rightarrow \infty} \frac{x^4 + 2x^3 + 3}{2x^4 - x + 2}$

(r) $\frac{1}{2}$

(D) $\lim_{x \rightarrow 2} \frac{\sqrt{3x^2 + 3x - 9} - \sqrt{3x + 3}}{x - 2}$

(s) 2

Sol. Answer : A(q), B(s), C(r), D(s)

(A) $\lim_{x \rightarrow 0} \frac{\cos x - \cos 3x}{x(\sin x - \sin 3x)}$

$$\lim_{x \rightarrow 0} \frac{2 \sin\left(\frac{3x+x}{2}\right) \sin\left(\frac{3x-x}{2}\right)}{x \cdot 2 \cos\left(\frac{x+3x}{2}\right) \cdot \sin\left(\frac{x-3x}{2}\right)} = \lim_{x \rightarrow 0} \frac{2 \sin 2x \cdot \sin x}{-x \cdot 2 \cos 2x \cdot \sin x} = -2$$

(B) $\lim_{x \rightarrow 0} \frac{\sin 4x}{\tan 2x} = \lim_{x \rightarrow 0} \frac{\left(\frac{\sin 4x}{4x}\right)(4x)}{\left(\frac{\tan 2x}{2x}\right)(2x)} = 2$

(C) $\lim_{x \rightarrow \infty} \frac{x^4 + 2x^3 + 3}{2x^4 - x + 2} = \lim_{x \rightarrow \infty} \frac{1 + \frac{2}{x} + \frac{3}{x^4}}{2 - \frac{1}{x^3} + \frac{2}{x^4}} = \frac{1+0+0}{2-0+0} = \frac{1}{2}$

(D) $\lim_{x \rightarrow 2} \frac{\sqrt{3x^2 + 3x - 9} - \sqrt{3x + 3}}{x - 2}$

$$= \lim_{x \rightarrow 2} \frac{(3x^2 + 3x - 9) - (3x + 3)}{(x - 2)(\sqrt{3x^2 + 3x - 9} + \sqrt{3x + 3})}$$

$$= \lim_{x \rightarrow 2} \frac{(3x^2 - 12)}{(x - 2)(3 + 3)}$$

$$= \lim_{x \rightarrow 2} \frac{3(x - 2)(x + 2)}{(x - 2)(6)} = \frac{3 \times 4}{6} = 2$$

2. Match the following.

Column-I

(A) $f(x) = \frac{1}{\sqrt{x-2}}$

(B) $f(x) = \frac{3x - \sin x}{x + \sin x}$

(C) $f(x) = x \sin \frac{\pi}{x} \quad f(0) = 0$

(D) $f(x) = \tan^{-1} \frac{1}{x}$

Column-II

(p) $\lim_{x \rightarrow 0} f(x) = 1$

(q) $\lim_{x \rightarrow 0} f(x) = 0$

(r) $\lim_{x \rightarrow \infty} f(x) = 0$

(s) $\lim_{x \rightarrow 0} f(x)$ does not exist

Sol. Answer : A(r), B(p), C(q), D(r, s)

(A) $\lim_{x \rightarrow \infty} \frac{1}{\sqrt{x-2}} = 0$

(B) $\lim_{x \rightarrow 0} \frac{3x - \sin x}{x + \sin x} = \lim_{x \rightarrow 0} \frac{3 - \cos x}{1 + \cos x} = \frac{3-1}{1+1} = \frac{2}{2} = 1$

(C) $\lim_{x \rightarrow 0} x \sin \left(\frac{\pi}{x} \right) = 0 \times (\text{number belong to } [-1, 1]) = 0$

(D) $\lim_{x \rightarrow 0} \tan^{-1} \left(\frac{1}{x} \right)$

RHL = $\lim_{h \rightarrow 0} \tan^{-1} \left(\frac{1}{h} \right) \rightarrow \frac{\pi}{2}$

LHL = $\lim_{h \rightarrow 0} \tan^{-1} \left(-\frac{1}{h} \right) \rightarrow -\frac{\pi}{2}$

Hence limit does not exist.

3. Match the following :

Column-I

(A) $\lim_{x \rightarrow \frac{\pi}{4}} (\sin 2x)^{\tan^2 2x}$

(B) $\lim_{x \rightarrow \infty} \left(\frac{2x-1}{2x+1} \right)^x$

(C) $\lim_{x \rightarrow \frac{\pi}{2}} (\tan x)^{\tan 2x}$

(D) $\lim_{x \rightarrow \frac{\pi}{4}} \tan 2x \tan \left(\frac{\pi}{4} - x \right)$

Column-II

(p) $\frac{1}{2}$

(q) $e^{-\frac{1}{2}}$

(r) e^{-1}

(s) 1

Sol. Answer : A(q), B(r), C(s), D(p)

$$\begin{aligned}
 \text{(A)} \quad \lim_{x \rightarrow \frac{\pi}{4}} (\sin 2x)^{\tan^2 2x} &= e^{\lim_{x \rightarrow \frac{\pi}{4}} (\sin 2x - 1) \cdot \tan^2 2x} \\
 &= e^{\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin 2x - 1}{\cot^2 2x}} = e^{\lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \cos 2x}{2 \cot 2x \cdot (-\operatorname{cosec}^2 2x) \times 2}} \\
 &= e^{\frac{1}{2}}
 \end{aligned}$$

$$\text{(B)} \quad \lim_{x \rightarrow \infty} \left(\frac{2x-1}{2x+1} \right)^x = e^{\lim_{x \rightarrow \infty} \left(\frac{2x-1}{2x+1} - 1 \right) x} = e^{\lim_{x \rightarrow \infty} \left(\frac{-2}{2x+1} \right) x} = e^{-1}$$

$$\begin{aligned}
 \text{(C)} \quad \lim_{x \rightarrow \frac{\pi}{2}} (\tan x)^{\tan 2x} &= e^{\lim_{x \rightarrow \frac{\pi}{2}} (\log \tan x) \cdot \tan 2x} \\
 &= e^{\lim_{x \rightarrow \frac{\pi}{2}} \frac{\log(\tan x)}{(\cot 2x)}} = e^{\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec^2 x}{(\tan x)} \cdot \frac{1}{-2 \operatorname{cosec}^2 2x}} = e^0 = 1
 \end{aligned}$$

$$\begin{aligned}
 \text{(D)} \quad \lim_{x \rightarrow \frac{\pi}{4}} (\tan 2x) \tan \left(\frac{\pi}{4} - x \right) \\
 = \lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{2 \tan x}{1 - \tan^2 x} \right) \cdot \left(\frac{1 - \tan x}{1 + \tan x} \right) = \lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \tan x}{(1 + \tan x)^2} = \frac{1}{2}
 \end{aligned}$$

4. Match the function in column I with their domain of definition in column II.

Column-I

Column-II

$$\text{(A)} \quad f(x) = \sin^{-1} \log_3 \left(\frac{x}{3} \right)$$

$$\text{(p)} \quad (-\infty, \infty)$$

$$\text{(B)} \quad f(x) = \sqrt{x^2 - 5x + 6} + \sqrt{2x + 8 - x^2}$$

$$\text{(q)} \quad [-2, 0) \cup (0, 1)$$

$$\text{(C)} \quad f(x) = \frac{1}{2 - \sin 3x}$$

$$\text{(r)} \quad [-2, 2] \cup [3, 4]$$

$$\text{(D)} \quad f(x) = \frac{1}{\log_{10} \sqrt{1-x}} + \sqrt{x+2}$$

$$\text{(s)} \quad [1, 9]$$

Sol. Answer : A(s), B(r), C(p), D(q)

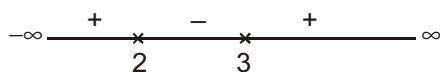
$$\text{(A)} \quad -1 \leq \log_3 \left(\frac{x}{3} \right) \leq 1$$

$$\Rightarrow 3^{-1} \leq \frac{x}{3} \leq 3^1$$

$$\Rightarrow 1 \leq x \leq 9$$

$$(B) \quad x^2 - 5x + 6 \geq 0$$

$$\Rightarrow (x-2)(x-3) \geq 0$$

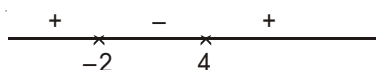


$$x \in [-\infty, 2] \cup [3, \infty) \quad \dots (i)$$

$$2x + 8 - x^2 \geq 0$$

$$x^2 - 2x - 8 \leq 0$$

$$(x-4)(x+2) \leq 0$$



$$\Rightarrow x \in [-2, 4] \quad \dots (ii)$$

By (i) and (ii)

$$x \in [-2, 2] \cup [3, 4]$$

$$(C) \quad f(x) = \frac{1}{2 - \sin 3x}$$

since $2 - \sin 3x$ is never equal to zero for any value of x . Hence domain $(f) = R$.

$$(D) \quad f(x) = \frac{1}{\log_{10} \sqrt{1-x}} + \sqrt{x+2}$$

For $f(x)$ to be real, we have

$$\sqrt{1-x} \neq 1, \quad 1-x > 0, \text{ i.e., } x \neq 0, \quad x < 1 \text{ and } x \geq -2$$

$$\Rightarrow x \in (-\infty, 1) - \{0\} \cap [-2, \infty)$$

$$\Rightarrow x \in [-2, 0) \cup (0, 1)$$

5. Match column I to column II according to the given conditions.

Column I

Column II

$$(A) \quad \lim_{x \rightarrow 0} \frac{(e^{\tan x} - e^{\sin x})}{(\tan x - \sin x)} \text{ equals}$$

$$(p) \quad 1$$

$$(B) \quad \lim_{x \rightarrow 0} \frac{\int_0^x x e^x dx}{x \sin x} \text{ equals}$$

$$(q) \quad \frac{1}{2}$$

$$(C) \quad \lim_{x \rightarrow -\infty} \frac{x+1}{\sqrt{4x^2+x+2}} \text{ equals}$$

$$(r) \quad -\frac{1}{2}$$

$$(D) \quad \lim_{n \rightarrow \infty} (\sin x)^{2n}, \text{ where } n \in N, \text{ is not equal to}$$

$$(s) \quad \frac{3}{4}$$

$$(t) \quad 2$$

Sol. Answer : A(p), B(q), C(r), D(q, r, s, t)

$$(A) \lim_{x \rightarrow 0} \frac{e^{\tan x} - e^{\sin x}}{(\tan x - \sin x)}$$

$$= \lim_{x \rightarrow 0} \frac{e^{\sin x} (e^{\tan x - \sin x} - 1)}{(\tan x - \sin x)} = e^0 \times 1 = 1$$

$$(B) \lim_{x \rightarrow 0} \frac{\int_0^x x e^x}{x \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{x e^x}{x \cos x + \sin x} \quad (\text{Using L - H Rule})$$

$$= \lim_{x \rightarrow 0} \frac{x e^x + e^x}{-x \sin x + \cos x + \cos x} = \frac{1}{2} \quad (\text{Using L - H Rule})$$

$$(C) \lim_{x \rightarrow -\infty} \frac{x+1}{|x| \sqrt{4 + \frac{1}{x} + \frac{2}{x^2}}}$$

$$= \lim_{x \rightarrow -\infty} \frac{x+1}{-x \sqrt{4 + \frac{1}{x} + \frac{2}{x^2}}} = -\frac{1}{2}$$

$$(D) \lim_{n \rightarrow \infty} (\sin x)^{2n}$$

Case (i) if $\sin x \neq (\pm 1)$

$$\Rightarrow \lim_{n \rightarrow \infty} (\sin x)^{2n} = 0$$

Case (ii) If $\sin x = \pm 1$

$$\Rightarrow \lim_{n \rightarrow \infty} (\pm 1)^{2n} = 1$$

6. Match column I to column II according to the right hand limit or left hand limit at the given point, where $[\]$ represents the greatest integer function.

Column I

$$(A) \lim_{x \rightarrow 0^-} \left[\frac{\sin[x]}{[x]} \right]$$

$$(B) \lim_{x \rightarrow 0^-} \left[\frac{\tan[x]}{[x]} \right]$$

$$(C) \lim_{x \rightarrow 10} [x]$$

$$(D) \lim_{x \rightarrow \frac{\pi}{2}} [\cos x]$$

Column II

(p) 0

(q) 10

(r) 9

(s) -1

(t) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos x} - 1}{\cos x}$

(u) 1

Sol. Answer : A(p), B(t, u), C(q, r), D (p, s)

$$(A) \text{ L H L} = \lim_{h \rightarrow 0} \left[\frac{\sin[0-h]}{[0-h]} \right] = [\sin 1] = 0$$

$$(B) \text{ L H L} = \lim_{h \rightarrow 0} \left[\frac{\tan[0-h]}{[0-h]} \right] = \left[\frac{\tan(-1)}{-1} \right] = [\tan 1] = 1$$

$$(C) \text{ R H L} = \lim_{h \rightarrow 0} [10+h] = 10$$

$$\text{L H L} = \lim_{h \rightarrow 0} [10-h] = 9$$

$$(D) \text{ R H L} = \lim_{h \rightarrow 0} \left[\cos\left(\frac{\pi}{2} + h\right) \right] = -1$$

$$\text{L H L} = \lim_{h \rightarrow 0} \left[\cos\left(\frac{\pi}{2} - h\right) \right] = 0$$

SECTION - F

Integer Answer Type Questions

1. If $y = f(x)$ and $y^4 - 4y + x = 0$. If $f(-8) = 2$ then the value of $|28 f'(-8)|$ is _____.

Sol. Answer (1)

$$y^4 - 4y + x = 0$$

$$\text{at } y = 2, x = 8$$

Differentiating the equation

$$4(y^3 - 1) \frac{dy}{dx} + 1 = 0$$

$$\frac{dy}{dx} = \frac{-1}{4(y^3 - 1)}$$

$$\Rightarrow f'(x) = -\frac{1}{4(y^3 - 1)}$$

$$f'(-8) = -\frac{1}{4(2^3 - 1)} = -\frac{1}{28}$$

$$\Rightarrow |28 f'(-8)| = 1$$

2. If $f(x) = f(1-x)$ and $f(x)$ is differentiable at every real value of x then the value of $f'\left(\frac{1}{2}\right) + f'\left(\frac{1}{4}\right) + f'\left(\frac{3}{4}\right)$ is ____.

Sol. Answer (0)

$$f(x) = f(1-x)$$

$$\Rightarrow f'(x) = -f'(1-x)$$

$$\text{at } x = \frac{1}{2}$$

$$f'\left(\frac{1}{2}\right) = -f'\left(1 - \frac{1}{2}\right)$$

$$f'\left(\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow f'\left(\frac{1}{2}\right) = 0 \quad \dots (i)$$

$$\text{Also } x = \frac{1}{4}$$

$$f'\left(\frac{1}{4}\right) = -f'\left(\frac{3}{4}\right)$$

$$\Rightarrow f'\left(\frac{1}{4}\right) + f'\left(\frac{3}{4}\right) = 0 \quad \dots (ii)$$

By (i) and (ii),

$$f'\left(\frac{1}{2}\right) + f'\left(\frac{1}{4}\right) + f'\left(\frac{3}{4}\right) = 0$$

3. If the value of $\lim_{x \rightarrow 0} (1 + \sin x + \sin^2 x)^{\cot x}$ is k then $\frac{2k}{e}$ is equal to ____.

Sol. Answer (2)

$$\lim_{x \rightarrow 0} (1 + \sin x + \sin^2 x)^{\cot x}$$

$$= e^{\lim_{x \rightarrow 0} (\sin x + \sin^2 x) \cdot \cot x}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{\sin x + \sin^2 x}{\tan x} \right)}$$

$$= e^{\lim_{x \rightarrow 0} \frac{x + x^2}{x}}$$

$$= e^1 = e$$

$$k = e \Rightarrow \frac{2k}{e} = \frac{2 \times e}{e} = 2$$

4. The value of $\lim_{x \rightarrow 0} \frac{e^{\sin^3 x} - \cos((\sin x)^3)}{x^3}$ is _____.

Sol. Answer (1)

$$\lim_{x \rightarrow 0} \frac{x^{\sin^3 x} - \cos((\sin x)^3)}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{x^{x^3} - \cos(x^3)}{x^3} \quad \left(\text{using } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right)$$

$$\lim_{x \rightarrow 0} \frac{(e^{x^3})(3x^2) + (\sin(x^3))(3x^2)}{(3x^2)} = \lim_{x \rightarrow 0} (e^{x^3} + (\sin x^3)) = e^0 + 0 = 1$$

5. Let $f(x)$ is a function satisfying the condition $f(x) = \frac{1}{3} \left[f(x+6) + \frac{6}{f(x+7)} \right]$ and $f(x) > 0 \forall x \in R$ then if $\lim_{x \rightarrow \infty} f(x) = \sqrt{m}$ then the value of m is _____.

Sol. Answer (3)

$$\therefore \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} f(x+6) = \lim_{x \rightarrow \infty} f(x+7) = m$$

then from the given condition

$$m = \frac{1}{3} \left[m + \frac{6}{m} \right]$$

$$\Rightarrow 3m^2 = m^2 + 6$$

$$\Rightarrow 2m^2 = 6$$

$$\Rightarrow m = \sqrt{3}$$

$$\text{so, } \lim_{x \rightarrow \infty} f(x) = \sqrt{3}$$

so the value of $m = 3$

6. If $\lim_{x \rightarrow 0} \frac{1}{x^2} (e^{2mx} - e^x - x) = \frac{3}{2}$ then the value of m is _____.

Sol. Answer (1)

$$\text{Given that } \lim_{x \rightarrow 0} \frac{1}{x^2} (e^{2mx} - e^x - x) = \frac{3}{2}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x(2m-2) + x^2 \left(2m^2 - \frac{1}{3} \right) + x^3(\dots) + \dots}{x^2} = \frac{3}{2}$$

$$\Rightarrow m = 1$$

7. If $\lim_{x \rightarrow 0} \left[\frac{m \sin x}{x} \right] = 8$ then the value of m is ($[.]$ in the greatest integer function).

Sol. Answer (9)

We know that

$$\frac{\sin x}{x} < 1$$

$$\Rightarrow \frac{m \sin x}{x} < m$$

$$\Rightarrow \lim_{x \rightarrow 0} \left[\frac{m \sin x}{x} \right] = (m-1)$$

$$\text{But given that } \lim_{x \rightarrow 0} \left[\frac{m \sin x}{x} \right] = 8$$

$$\text{so } m-1 = 8$$

$$\Rightarrow m = 9$$

8. The largest value of the non-negative integer a for which $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$ is

[JEE(Advanced)-2014]

Sol. Answer (0)

$$\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}}$$

$$\lim_{x \rightarrow 1} \left\{ \frac{a(1-x) + \sin(x-1)}{(x-1) + \sin(x-1)} \right\}^{1+\sqrt{x}}$$

$$\lim_{x \rightarrow 1} \left\{ \frac{-a + \frac{\sin(x-1)}{x-1}}{1 + \frac{\sin(x-1)}{x-1}} \right\}^{(1+\sqrt{x})}$$

$$\left\{ \frac{-a+1}{2} \right\}^2 = \frac{1}{4}$$

$$\frac{-a+1}{2} = \pm \frac{1}{2}$$

$$-a+1 = \pm 1 \Rightarrow a = 0, a = 2$$

But at $a = 2$, base becomes negative

$$\therefore a = 0$$

9. Let m and n be two positive integers greater than 1. If $\lim_{\alpha \rightarrow 0} \left(\frac{e^{\cos(\alpha^n)} - e}{\alpha^m} \right) = -\left(\frac{e}{2}\right)$ then the value of $\frac{m}{n}$ is

[JEE(Advanced)-2015]

Sol. Answer (2)

$$\lim_{\alpha \rightarrow 0} \frac{e^{\cos \alpha^n} - e}{\alpha^m} = -\frac{e}{2}$$

 $\Rightarrow$ Using L' Hospital's rule

$$\lim_{\alpha \rightarrow 0} \frac{e^{\cos(\alpha^n)} \cdot -\sin(\alpha^n) \cdot n\alpha^{n-1}}{m \cdot \alpha^{m-1}} = \frac{-e}{2}$$

$$\Rightarrow \boxed{\frac{m}{n} = 2}$$

10. Let $\alpha, \beta \in \mathbb{R}$ be such that $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals

[JEE(Advanced)-2016]

Sol. Answer (7)

$$\lim_{x \rightarrow 0} \frac{x^2 \left(\beta x - \frac{(\beta x)^3}{3!} + \dots \right)}{\alpha x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \right)}$$

$$\boxed{\alpha = 1} \quad \frac{\beta}{\frac{1}{3!}} = 1$$

$$\beta = \frac{1}{6}$$

$$6 \left(1 + \frac{1}{6} \right) = 7$$

SECTION - G**Multiple True-False Type Questions**

1. STATEMENT-1 : $\lim_{x \rightarrow \infty} \frac{\log[x]}{[x]} = 0$, where $[x]$ represents the integral part of x .

STATEMENT-2 : $\lim_{x \rightarrow 0} \frac{\sqrt{\sec^2 x - 1}}{x}$ does not exist.

STATEMENT-3 : $\lim_{x \rightarrow 2} (x-1)^{\frac{1}{x-2}} = 1$

(1) T T T

(2) T T F

(3) F T F

(4) F F F

Sol. Answer (2)

2. $\lim_{x \rightarrow 0} \frac{A \sin x + B \log(1+x^2) + C(1-\cos x)}{x^2} = 2$, then

STATEMENT-1 : $A = 1$

STATEMENT-2 : $2B + C = 4$

STATEMENT-3 : $A + B + C = 0$

(1) T T T

(2) T T F

(3) F T F

(4) F F F

Sol. Answer (3)

$$\lim_{x \rightarrow 0} \frac{A \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \right) + B \left(x^2 - \frac{x^4}{2} + \dots \right) + C \left(1 - \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) \right)}{x^2} = 2$$

$$\lim_{x \rightarrow 0} \frac{x(A) + x^2 \left(B + \frac{C}{2} \right) + x^3 \left(-\frac{A}{3!} \right) + \dots}{x^2} = 2$$

$$\Rightarrow A = 0, B + \frac{C}{2} = 2$$

SECTION - H

Aakash Challengers Questions

1. If $\lim_{x \rightarrow 0} \frac{x^3}{\sqrt{a+x}(bx - \sin x)} = 1$, $a \in \mathbb{R}^+$, then the value of $a + b + 1975$ is

Sol. Given $\lim_{x \rightarrow 0} \frac{x^3}{\sqrt{a} \left(1 + \frac{x}{a} \right)^{\frac{1}{2}} (bx - \sin x)} = 1$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^3}{\sqrt{a} \left\{ 1 + \frac{1}{2} \left(\frac{x}{a} \right) + \frac{1}{2} \left(-\frac{1}{2} \right) \left(\frac{x}{a} \right)^2 + \dots \right\} \left\{ x(b-1) + \frac{x^3}{3!} + \dots \right\}}$$

$$\Rightarrow \frac{3!}{\sqrt{a}} = 1 \text{ and } b = 1$$

$$\Rightarrow a = 36, b = 1$$

$$\therefore a + b + 1975 = 36 + 1 + 1975 = 2012$$

2. Evaluate; $\lim_{n \rightarrow \infty} \left(\frac{1 + \sqrt[n]{4}}{2} \right)^n$.

Sol. $\lim_{n \rightarrow \infty} \left(\frac{1 + \sqrt[n]{4}}{2} \right)^n = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{4} - 1}{2} \cdot n = e^{\lim_{n \rightarrow \infty} \frac{1}{2} \cdot \frac{\frac{1}{4^n} - 1}{1/n}} = e^{\frac{1}{2} \log 4} = e \log \sqrt{4} = \sqrt{4} = 2$

3. If $\lim_{x \rightarrow 1} (1 + ax + bx^2)^{\frac{c}{x-1}} = e^3$, then the value of $a + b + bc + 2009$ is

Sol. Given $\lim_{x \rightarrow 1} (1 + ax + bx^2)^{\frac{c}{x-1}} = e^3$

$$\Rightarrow e^{\lim_{x \rightarrow 1} \frac{(ax + bx^2)c}{x-1}} = e^3 \quad (\because a + b = 0)$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{(ax + bx^2)c}{x-1} = 3$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{(a + 2b)c}{1} = 3$$

$$\Rightarrow ac + 2bc = 3$$

$$\Rightarrow ac + bc + bc = 3$$

$$\Rightarrow c(a + b) + bc = 3$$

$$\Rightarrow bc = 3 \quad (\because a + b = 0)$$

$$\therefore a + b + bc + 2009$$

$$= 3 + 2009$$

$$= 2012$$

4. If $\lim_{x \rightarrow 0} \frac{\sin 2x + a \sin x}{x^3} = b$ (finite), then $(ab)^2$ equals

Sol. $\lim_{x \rightarrow 0} \frac{\sin 2x + a \sin x}{x^3} = b$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\left(2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} + \dots\right) + a\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots\right)}{x^3} = b$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x(a+2) + x^3\left(-\frac{8}{6} - \frac{a}{6}\right) + x^5(\dots) + \dots}{x^3} = b$$

$$\Rightarrow a + 2 = 0 \quad \Rightarrow a = -2$$

$$-\frac{8}{6} - \frac{a}{6} = b \quad \Rightarrow b = -1$$

$$\Rightarrow (ab)^2 = 2^2 = 4.$$

5. Find the value of $\lim_{x \rightarrow \infty} 48x \left(\frac{\pi}{4} - \tan^{-1} \frac{x+1}{x+2} \right)$ if $\tan^{-1} \tan \theta = \theta$ if $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

Sol. $\lim_{x \rightarrow \infty} 48x \left(\frac{\pi}{4} - \tan^{-1} \frac{x+1}{x+2} \right)$

$$= \lim_{x \rightarrow \infty} \frac{48 \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{x+1}{x+2} \right) \right)}{\left(\frac{1}{x} \right)}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow \infty} \frac{48 \left(\frac{1}{1 + \left(\frac{x+1}{x+2} \right)^2} \cdot \left(\frac{1}{(x+2)^2} \right) \right)}{\left(-\frac{1}{x^2} \right)} \\
 &= \lim_{x \rightarrow \infty} 48 \left(\frac{x^2}{(x+2)^2 + (x+1)^2} \right) \\
 &= 48 \times \frac{1}{2} \\
 &= 24
 \end{aligned}$$

6. Let $y = 1 + \frac{a_1}{x-a_1} + \frac{a_2 x}{(x-a_1)(x-a_2)} + \frac{a_3 x^2}{(x-a_1)(x-a_2)(x-a_3)} + \dots + \frac{a_n x^{n-1}}{(x-a_1)(x-a_2)(x-a_3)\dots(x-a_n)}$. Find $\frac{dy}{dx}$.

Sol. $y = \left(1 + \frac{a_1}{x-a_1} \right) + \frac{a_2 x}{(x-a_1)(x-a_2)} + \dots$

$$y = \left(\frac{x}{x-a_1} + \frac{a_2 x}{(x-a_1)(x-a_2)} \right) + \dots$$

$$y = \frac{x}{x-a_1} \left(1 + \frac{a_2}{x-a_2} \right) + \frac{a_3 x^2}{(x-a_1)(x-a_2)(x-a_3)} + \dots$$

$$y = \frac{x^2}{(x-a_1)(x-a_2)} + \frac{a_3 x^2}{(x-a_1)(x-a_2)(x-a_3)} + \dots$$

$$y = \frac{x^3}{(x-a_1)(x-a_2)(x-a_3)} + \dots \text{ and so on.}$$

Hence y can be sum use as

$$y = \frac{x^n}{(x-a_1)(x-a_2)\dots(x-a_n)}$$

$$\log y = n \log x - \log(x-a_1) - \log(x-a_2) - \log(x-a_3) + \dots$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{n}{x} - \frac{1}{x-a_1} - \frac{1}{x-a_2} \dots$$

$$= \left(\frac{1}{x} - \frac{1}{x-a_1} \right) + \left(\frac{1}{x} - \frac{1}{x-a_2} \right) + \dots$$

$$= \frac{-a_1}{x(x-a_1)} + \frac{-a_2}{x(x-a_2)} + \dots$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} \left[\frac{a_1}{x-a_1} + \frac{a_2}{x-a_2} + \dots \right]$$

7. If $y = \frac{1}{1+x^{a-b}+x^{c-b}} + \frac{1}{1+x^{b-c}+x^{a-c}} + \frac{1}{1+x^{b-a}+x^{c-a}}$ then find $\frac{dy}{dx}$ at e^{abc} .

Sol. $y = \frac{1}{1+x^{a-b}+x^{c-b}} + \frac{1}{1+x^{b-c}+x^{a-c}} + \dots$

$$y = \frac{x^b}{x^a+x^b+x^c} + \frac{x^c}{x^a+x^b+x^c} + \frac{x^a}{x^a+x^b+x^c}$$

$$y = \frac{x^a+x^b+x^c}{x^a+x^b+x^c} = 1$$

$$\frac{dy}{dx} = 0$$

8. Find the number of integral(s) of the equation $[x] + [2x] + [4x] + [8x] + \dots + [32x] = 12345$. Where $[]$ denotes the greatest integer function.

Sol. $12345 = [x] + [2x] + [4x] + [8x] + [16x] + [32x]$

$$12345 \leq x + 2x + 4x + 8x + 16x + 32x$$

$$12345 \leq 63x$$

$$\Rightarrow x \geq \frac{12345}{63} = 195 \frac{20}{21}$$

When $x = 196$

The LHS of given equation is 12348.

$$\therefore 195 \frac{20}{21} \leq x < 196$$

Now consider the interval $\left[195 \frac{20}{21}, 196\right)$

$$\begin{aligned} [x] + [2x] + [4x] + [8x] + [16x] + [32x] \\ = 195 + 391 + 783 + 1567 + 3135 + 6271 \\ = 12342 < 12345 \end{aligned}$$

The LHS is less than 12342

$\therefore$ For no value of x , then given equality will hold

9. If $\lim_{x \rightarrow 0} f(x)$ exists and is equal to p , then find p and q where, $f(x) = \begin{cases} (\sin x + \cos x)^{\operatorname{cosec} x} & , -\frac{\pi}{2} < x < 0 \\ \frac{e^{\frac{1}{x}} + e^{\frac{2}{x}} + e^{\frac{3}{x}}}{pe^{-2+\frac{1}{x}} + qe^{-1+\frac{3}{x}}} & , 0 < x < \frac{\pi}{2} \end{cases}$

Sol. $f(x) = \begin{cases} (\sin x + \cos x)^{\operatorname{cosec} x} & ; -\frac{\pi}{2} < x < 0 \\ \frac{e^{\frac{1}{x}} + e^{\frac{2}{x}} + e^{\frac{3}{x}}}{pe^{-2+\frac{1}{x}} + qe^{-1+\frac{3}{x}}} & ; 0 < x < \frac{\pi}{2} \end{cases}$

$$f(0^+) = \lim_{h \rightarrow 0} \frac{e^{\frac{1}{h}} + e^{\frac{2}{h}} + e^{\frac{3}{h}}}{pe^{-2+\frac{1}{h}} + qe^{-1+\frac{3}{h}}} = \lim_{h \rightarrow 0} \frac{1 + e^{\frac{1}{h}} + e^{\frac{2}{h}}}{pe^{-2} + qe^{-1}} = \lim_{h \rightarrow 0} \frac{4e^{\frac{2}{h}} + e^{\frac{1}{h}} + 1}{pe^{-2}e^{\frac{2}{h}} + qe^{-1}e^{\frac{2}{h}}} = \frac{1}{qe^{-1}} = \frac{e}{q}$$

$$\begin{aligned}
 f(0^-) &= \lim_{h \rightarrow 0} (\sin(0-h) + \cos(0-h))^{\operatorname{cosec}(0-h)} \\
 &= \lim_{h \rightarrow 0} (\cos h - \sin h)^{-\operatorname{cosec} h} \\
 &= e^{\lim_{h \rightarrow 0} (\cos h - \sin h) \times \frac{1}{(-\sin h)}} \\
 &= e^{\lim_{h \rightarrow 0} \frac{1 + \sin h - \cos h}{\sin h}} \\
 &= e^{\lim_{h \rightarrow 0} \frac{\cos h + \sin h}{\cos h}} = e
 \end{aligned}$$

$$\text{Put } f(0^+) = f(0^-)$$

$$\Rightarrow e = \frac{e}{q}$$

$$\Rightarrow q = 1 \text{ and } p \in R$$

10. Evaluate $\lim_{x \rightarrow 0} \frac{k^{|x|} - 1 - |x| \ln k}{x^2}$, $k > 0$.

Sol. $\lim_{x \rightarrow 0} \frac{k^{|x|} - 1 - |x| \ln k}{x^2}$, here we have to find RHL and LHL.

$$\begin{aligned}
 \text{RHL} &= \lim_{x \rightarrow 0^+} \frac{k^{|x|} - 1 - |x| \ln k}{x^2} = \lim_{h \rightarrow 0} \frac{k^{|h|} - 1 - |h| \ln k}{h^2} \\
 &= \lim_{h \rightarrow 0} \frac{k^h - 1 - h \ln k}{h^2} = \lim_{h \rightarrow 0} \frac{k^h \log k - \log k}{2h} \\
 &= \lim_{h \rightarrow 0} \frac{k^h (\log k)^2}{2} = \frac{1}{2} (\log k)^2
 \end{aligned}$$

$$\text{LHL} = \lim_{h \rightarrow 0} \frac{k^{|0-h|} - 1 - |0-h| \ln k}{h^2} = \frac{1}{2} (\log k)^2$$

